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Quantum-inspired robust optimization for coordinated scheduling of PV-hydrogen microgrids under multi-dimensional uncertainties

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The integration of photovoltaic (PV) generation and hydrogen storage in rural microgrids enables clean, long-duration energy supply, yet introduces operational challenges under high uncertainty. These include fluctuations in PV output, stochastic hydrogen demand, and volatile market prices. Traditional deterministic or static robust methods often fall short in handling such dynamic, multi-dimensional uncertainties. To address these gaps, this study develops a Quantum-Inspired Robust Optimization (QRO) framework that coordinates PV-H_2 microgrid scheduling by dynamically adapting to real-time disturbances. The QRO approach leverages quantum-classical hybrid principles and integrates distributionally robust optimization with reinforcement learning. Uncertainty sets evolve adaptively based on operational feedback rather than remaining fixed, enhancing resilience to cyberattacks, extreme weather, and grid outages. Deep Q-learning and policy gradient methods are employed to continuously refine dispatch strategies, ensuring robust performance in non-stationary environments. A case study on a 5 MW PV-H_2 microgrid with a 3 MW electrolyzer and 2 MW fuel cell demonstrates the practical effectiveness of the proposed framework. The model incorporates real historical solar profiles, stochastic demand, and price signals over a full-year horizon. Stress-testing under scenarios such as 48-hour grid failure, signal-based cyberattacks, and 40% PV output curtailment reveals substantial gains: operational costs are reduced by 9.3%, resilience scores improve by over 20% in adverse conditions, and convergence speed increases by 42% relative to classical optimization. These improvements reflect not only enhanced adaptability and computational efficiency, but also the practical feasibility of real-time learning in resilient energy scheduling. Importantly, the term "quantum-inspired" refers to classical algorithms that emulate quantum principles—such as probabilistic reasoning and solution diversity—without employing quantum hardware. By unifying quantum-inspired modeling, distributional robustness, and reinforcement learning, this framework offers a scalable and adaptive solution for next-generation hydrogen-based microgrid operations.

The integration of renewable energy into modern power grids has been a central focus in the pursuit of low-carbon energy systems. Among various renewable energy sources, photovoltaic systems have emerged as a highly scalable, economically viable, and widely deployable technology. However, PV generation is inherently intermittent, influenced by weather conditions, diurnal variations, and seasonal fluctuations, which pose significant challenges to the stability and reliability of power distribution networks[1,2]. To address the intermittency issue, hydrogen energy storage systems have gained increasing attention as a viable long-term energy carrier. By leveraging electrolysis, excess electricity from PV generation can be converted into green hydrogen, which can be stored and later reconverted into electricity using fuel cells[3,4]. This approach enables seasonal energy storage, enhances grid flexibility, and supports deep decarbonization of energy-intensive sectors such as transportation, industrial heating, and power backup systems.

Despite the promising benefits of PV-H_2 microgrids, the optimal coordination of generation, electrolysis, storage, and dispatch remains a complex challenge. The uncertainties associated with PV generation, hydrogen demand, storage losses, and energy market prices introduce significant operational risks[5]. Traditional

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deterministic optimization methods fail to adequately capture these uncertainties, leading to suboptimal and potentially unreliable energy management strategies. Moreover, classical stochastic and robust optimization methods rely on predefined probabilistic distributions or worst-case assumptions, which often result in overly conservative decisions that fail to exploit the full potential of available resources[6–8]. To overcome these limitations, there is an urgent need for an advanced optimization framework that can dynamically adapt to real-time uncertainties, ensuring optimal scheduling of energy resources under unpredictable conditions.

This paper aims to develop a novel quantum-inspired robust optimization framework for optimizing the operation of a PV-H_2 microgrid, integrating photovoltaic generation, hydrogen electrolysis, hydrogen storage, and fuel cell dispatch. The primary objectives of this research are to develop a quantum-enhanced energy scheduling model that efficiently handles uncertainty in PV generation and hydrogen demand, optimize the coordinated operation of electrolysis, hydrogen storage, and fuel cell dispatch to minimize total system costs while ensuring energy security, leverage reinforcement learning and quantum computing principles to enhance decision-making adaptability under stochastic conditions, and ensure resilience against extreme weather events, cyber threats, and grid failures by integrating adaptive risk-mitigation strategies. By addressing these objectives, this study provides a groundbreaking optimization framework that outperforms conventional deterministic and stochastic methods, offering greater adaptability, computational efficiency, and resilience in PV-H_2 microgrid operations.

This paper integrates hybrid quantum-classical computing principles, where quantum-assisted optimization is used to solve non-convex, multi-stage dispatch problems while ensuring robust constraint satisfaction. By combining quantum variational algorithms with distributionally robust optimization, the model dynamically adapts to hydrogen market fluctuations, operational constraints, and system failures. To enhance real-time decision-making, this work incorporates deep Q-learning and policy gradient reinforcement learning to iteratively update dispatch policies based on system feedback. This self-learning mechanism ensures that the microgrid can proactively adjust its operations, maximizing efficiency while minimizing cost. This research develops a multi-layered resilience framework that integrates stochastic risk modeling, uncertainty propagation, and cyber-physical risk assessments. The model accounts for extreme event scenarios, ensuring that the microgrid remains operational under critical contingencies such as grid failures, natural disasters, and cyberattacks. By introducing these state-of-the-art techniques, this paper presents a comprehensive, scalable, and computationally efficient framework for optimizing PV-H_2 microgrid operations under real-world uncertainty.

This paper makes several key contributions. First, it proposes a quantum-inspired robust optimization framework for PV-H_2 microgrid scheduling, integrating quantum-enhanced uncertainty modeling, hybrid quantum-classical computing, and distributionally robust optimization techniques. Second, it develops a reinforcement learning-based adaptive energy scheduling model, ensuring that hydrogen storage and dispatch strategies dynamically evolve based on real-time operational feedback. Third, it introduces a hybrid quantum-classical optimization methodology, leveraging quantum variational algorithms and deep reinforcement learning to optimize long-term energy storage decisions.

Literature review

Traditional optimization approaches for renewable energy systems can be broadly categorized into deterministic optimization, stochastic optimization, robust optimization, and artificial intelligence-based methods. Deterministic optimization techniques, such as mixed-integer linear programming (MILP) and nonlinear programming (NLP), have been widely applied for microgrid scheduling, focusing on cost minimization, emissions reduction, and reliability enhancement[9,10]. However, these methods assume that all parameters are known with certainty, which is unrealistic for PV-H_2 systems, given the inherent variability in solar generation and hydrogen demand. Stochastic optimization has been introduced to account for uncertainty by modeling uncertain parameters using probability distributions. Scenario-based stochastic programming and chance-constrained programming have been applied to optimize microgrid operations under uncertain conditions[11,12]. Although stochastic optimization improves decision-making under uncertainty, its performance heavily depends on the accuracy of probability distributions, which are often difficult to estimate accurately. Furthermore, solving large-scale stochastic problems requires extensive computational resources, making them less practical for real-time microgrid operation.

Robust optimization has emerged as an alternative to stochastic programming, providing a worst-case optimization framework that ensures feasibility under the most extreme uncertainty realizations. Distributionally robust optimization (DRO) further extends this approach by allowing the uncertainty set to be characterized by a family of probability distributions rather than a single predefined distribution[13,14]. Several studies have applied robust and DRO techniques to microgrid optimization, demonstrating their ability to enhance resilience against uncertainty. However, these methods often lead to overly conservative solutions, reducing the utilization of available resources and increasing operational costs. Artificial intelligence (AI) and machine learning-based optimization methods have also been explored for microgrid scheduling. Reinforcement learning (RL) has gained popularity as a model-free approach that can learn optimal policies through interaction with the environment. Deep Q-Networks (DQN), policy gradient methods, and Proximal Policy Optimization (PPO) have been applied to optimize microgrid energy management[15,16]. These techniques offer adaptability and can improve decision-making under dynamic conditions, but their reliance on extensive training data and computational complexity pose challenges for real-time implementation.

Hydrogen energy storage has been widely recognized as a promising solution for addressing the intermittency of renewable energy sources and enhancing grid resilience. Several studies have focused on optimizing the operation of hydrogen-based microgrids, incorporating electrolysis, storage, and fuel cell dispatch[17,18]. The majority of these studies employ either deterministic or stochastic optimization techniques to minimize operational costs and maximize energy efficiency. One branch of research focuses on optimizing hydrogen

production from surplus renewable energy. These studies primarily utilize MILP, NLP, or heuristic algorithms such as genetic algorithms (GA) and particle swarm optimization (PSO) to determine the optimal scheduling of electrolysis units. While these approaches can provide near-optimal solutions, they often fail to adequately address uncertainty in renewable generation and hydrogen demand, leading to suboptimal operational decisions[19,20]. Other studies have explored the integration of hydrogen storage into microgrids, aiming to balance supply and demand while minimizing hydrogen leakage and degradation losses. Dynamic programming and robust optimization approaches have been applied to optimize hydrogen storage and dispatch decisions under uncertainty. However, these methods often assume predefined uncertainty sets, limiting their ability to adapt to real-time system variations[21]. More recent works have investigated the role of hydrogen in multi-energy systems, integrating hydrogen production, storage, and fuel cells with electricity and thermal networks[22]. These studies emphasize the need for coordinated operation of multiple energy carriers, but they generally rely on conventional optimization methods that do not fully exploit the potential of AI-driven and quantum-inspired approaches.

Quantum computing has recently emerged as a powerful tool for solving complex optimization problems that are intractable for classical methods. Quantum-inspired optimization algorithms, including quantum annealing and variational quantum eigensolvers, have been applied to combinatorial and high-dimensional optimization problems in various fields. In the energy sector, quantum computing has shown promise for unit commitment, demand response, and energy trading optimization[23,24]. However, its application to PV-H_2 microgrids remains largely unexplored. Quantum-inspired robust optimization (QRO) provides a novel approach to handling uncertainty by leveraging quantum probability distributions and superposition principles. Unlike classical stochastic and robust optimization, which rely on fixed probability distributions or worst-case scenarios, QRO enables dynamic adjustment of uncertainty sets based on real-time information. By exploring multiple solution paths simultaneously, quantum annealing-based optimization reduces computational complexity while achieving near-optimal solutions[25]. Several studies have demonstrated the potential of hybrid quantum-classical algorithms for solving energy management problems[26]. For instance, quantum-enhanced reinforcement learning has been proposed for demand-side management, where quantum variational algorithms are used to accelerate the learning process[27,23]. These hybrid approaches combine the strengths of quantum computing and AI to improve decision-making under uncertainty[28,29]. Despite these advancements, the application of quantum-inspired methods to microgrid optimization remains an emerging research area[10]. Most existing studies focus on theoretical formulations rather than practical implementations[30]. Moreover, there is a lack of comprehensive frameworks that integrate QRO with AI-driven adaptive scheduling, particularly for PV-H_2 systems. This research aims to bridge this gap by developing a hybrid quantum-classical optimization framework that leverages QRO and reinforcement learning for resilient microgrid scheduling.

Mathematical modeling and methodology

To mathematically formalize the operational scheduling and uncertainty management of the photovoltaic-hydrogen microgrid, this section develops a comprehensive optimization model that integrates cost minimization, operational feasibility, uncertainty propagation, and resilience enhancement. The model captures the full-chain energy flow from photovoltaic generation, electrolysis, hydrogen storage, and fuel cell dispatch to grid interactions and hydrogen market participation. Through a series of coupled mathematical equations, the developed model systematically describes energy conversion efficiencies, hydrogen storage dynamics, economic objectives, physical constraints, and risk mitigation strategies under multiple uncertainty sources. Each equation is constructed to directly link the physical and economic processes with the uncertainty-aware scheduling decisions, ensuring the model is capable of handling both normal operating conditions and extreme disturbance scenarios.

To solve the proposed optimization model efficiently while ensuring adaptability to real-time uncertainty and evolving operational environments, this section also introduces a novel hybrid methodology that combines quantum-inspired robust optimization with reinforcement learning-enhanced adaptive scheduling. It should be noted that the proposed framework is entirely implemented on classical computing platforms. The use of the term "quantum-inspired" throughout this paper reflects a heuristic modeling perspective rather than the application of actual quantum algorithms or hardware-based quantum computation. The methodological framework consists of three core layers: the quantum-enhanced uncertainty modeling layer, the distributionally robust optimization solver layer, and the reinforcement learning-based adaptive decision-making layer. The quantum-inspired layer leverages the superposition principle to capture probabilistic uncertainties in photovoltaic generation and hydrogen demand, enabling multi-scenario evaluation within a single optimization pass. The robust optimization solver constructs distributionally robust constraints to ensure feasibility under all plausible uncertainty distributions, while the reinforcement learning layer continuously updates dispatch policies based on real-time system feedback, enhancing operational flexibility. This hybrid methodology seamlessly integrates mathematical modeling and data-driven learning, ensuring that the microgrid's scheduling decisions evolve intelligently with changing external conditions and internal states. In this context, the term "adaptive uncertainty modeling" refers to the adjustment of scenario probabilities in the uncertainty ensemble based on observed deviations in photovoltaic generation and hydrogen demand. This probabilistic update is conceptually embedded within the optimization layer and operates independently from the reinforcement learning mechanism, which functions at the policy level.

To clarify the structure and workflow of the proposed methodology, a conceptual framework is provided to illustrate the interaction among core components. Given the integration of quantum-inspired uncertainty modeling, robust optimization, and reinforcement learning, it is important to visualize how these modules communicate and contribute to the overall scheduling process. The following diagram summarizes the key information flow and operational logic that underpin the adaptive control of the PV-H_2 microgrid under uncertainty (Fig. 1).

As shown in the figure, the PV-H_2 microgrid generates real-time operational data—such as renewable output and hydrogen flow rates—which feed into the scenario generation module. This module constructs quantum-inspired uncertainty sets that evolve dynamically with system states. These adaptive distributions are processed by the robust optimization engine to compute dispatch decisions that minimize cost while ensuring system resilience. Reinforcement learning agents provide continuous feedback to update both the uncertainty modeling and the dispatch policy, enabling closed-loop adaptability. Through this architecture, the framework achieves real-time learning, scenario diversity, and robust performance in complex and uncertain operating environments.

To ensure clarity in the mathematical formulation, all variables and parameters used in this study are defined in the following nomenclature table.

Symbol	Description	Unit
$\mathscr{G} = (\mathscr{N}, \mathscr{E})$	Graph representing the power network topology	–
$\mathscr{N}$	Set of buses/nodes in the network	–
$\mathscr{E}$	Set of edges/lines in the network	–
δ_ℓ	Binary decision variable for building line ℓ	-
$\varphi_\ell^{\mathrm{inv}}$	Investment cost of line ℓ	$
$\mathscr{L}^{\mathrm{op}}$	Operational loss function under scenario ω	$
ω	A renewable generation scenario	–
$G_{\boldsymbol{\theta}_G}$	GAN-based scenario generator with parameter $\boldsymbol{\theta}_G$	–
$\widehat{\mathscr{X}}^\star$	Optimal transmission expansion plan under worst-case scenario	–
$\mathscr{O}$	Operational decision variables set	–
c_i^{ENS}	Cost of energy not served at bus i	$ / kWh
P_i^{ENS}	Energy not served at bus i	kWh
$\mathrm{P}^{\mathrm{gen}}$	Power generation vector across buses	MW
$\mathrm{P}^{\mathrm{load}}$	Power demand vector across buses	MW
f_ℓ	Power flow on line ℓ	MW
$\overline{f}_\ell$	Thermal limit on line ℓ	MW
$\boldsymbol{\theta}$	Vector of bus voltage phase angles	rad
B_ℓ	Susceptance of line ℓ	p.u.
γ	Scaling parameter in quantum-inspired update rule	–
$\mathscr{D}$	Quantum dual update vector (inspired)	–
x_t	State vector at time t in reinforcement learning	–
a_t	Action taken at time t	–
r_t	Reward received at time t	–
$Q(\mathrm{x}, \mathrm{a})$	Q-value function in RL	–
$\pi(\mathrm{a}\|\mathrm{x})$	Policy function in RL	–

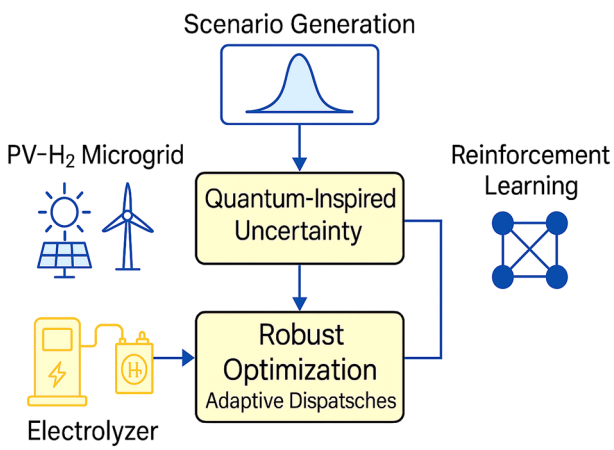

Fig. 1. QRO-RL framework for PV-H_2 microgrid scheduling.

Symbol	Description	Unit
$\mathscr{T}$	Transition probability in MDP	–
η	Learning rate in RL update	–
ϵ	Exploration parameter in ϵ-greedy policy	–

'–' represents no unit or non-numerical unit

$$\min_{\mathbf{\Phi},\mathbf{\Psi},\mathbf{\Lambda}} \sum_{t=1}^{T} \sum_{\iota\in\mathscr{N}} \left(\kappa_{\iota,t}^{\mathrm{PV}} \cdot \Phi_{\iota,t}^{\mathrm{PV}} + \kappa_{\iota,t}^{\mathrm{el}} \cdot \Psi_{\iota,t}^{\mathrm{el}} + \kappa_{\iota,t}^{\mathrm{st}} \cdot \Lambda_{\iota,t}^{\mathrm{H_2}}\right) + \sum_{\tau=1}^{T} \sum_{\vartheta\in\mathscr{M}} \left(\kappa_{\vartheta,\tau}^{\mathrm{fc}} \cdot \Lambda_{\vartheta,\tau}^{\mathrm{H_2}} + \kappa_{\vartheta,\tau}^{\mathrm{grid}} \cdot \Phi_{\vartheta,\tau}^{\mathrm{grid}}\right) \tag{1}$$

Equation (1) represents the total cost minimization function for the PV-H_2 microgrid. It considers the operational costs of PV generation $\Phi_{\iota,t}^{\mathrm{PV}}$, electrolyzer operation $\Psi_{\iota,t}^{\mathrm{el}}$, and hydrogen storage $\Lambda_{\iota,t}^{\mathrm{H_2}}$. Additionally, it includes costs related to the fuel cell system and grid transactions. The objective is to simultaneously optimize the trade-offs between renewable generation, hydrogen production, storage efficiency, and grid interaction. The summation over all nodes ι and all time steps t ensures a comprehensive cost evaluation for the entire operational horizon.

$$\max_{\mathbf{\Xi},\mathbf{\Omega}} \sum_{t=1}^{T} \sum_{\kappa\in\mathscr{K}} \left(\theta_{\kappa,t}^{\mathrm{H_2}} \cdot \Xi_{\kappa,t}^{\mathrm{sale}} + \vartheta_{\kappa,t}^{\mathrm{grid}} \cdot \Omega_{\kappa,t}^{\mathrm{trade}}\right) - \sum_{\tau=1}^{T} \sum_{\nu\in\mathscr{L}} \left(\epsilon_{\nu,\tau}^{\mathrm{leak}} \cdot \Lambda_{\nu,\tau}^{\mathrm{H_2}}\right) \tag{2}$$

Equation (2) models the maximization of revenue from hydrogen sales and grid electricity trading while accounting for hydrogen leakage losses. The first term in the equation captures hydrogen sales $\Xi_{\kappa,t}^{\mathrm{sale}}$, multiplied by the market price $\theta_{\kappa,t}^{\mathrm{H_2}}$ at each time step t. The second term represents grid electricity transactions $\Omega_{\kappa,t}^{\mathrm{trade}}$, influenced by dynamic electricity pricing $\vartheta_{\kappa,t}^{\mathrm{grid}}$. The last term introduces hydrogen leakage effects, where $\epsilon_{\nu,\tau}^{\mathrm{leak}}$ represents loss coefficients that capture storage inefficiencies over time.

$$\Lambda_{\iota,t}^{\mathrm{H_2}} = \alpha_{\iota,t} \cdot \Psi_{\iota,t}^{\mathrm{el}} - \beta_{\iota,t} \cdot \Lambda_{\iota,t}^{\mathrm{H_2}} + \gamma_{\iota,t} \cdot \Lambda_{\iota,t}^{\mathrm{fc}} + \sum_{\tau=1}^{t} \sum_{\kappa\in\mathscr{K}} \left(\varphi_{\kappa,\tau}^{\mathrm{res}} \cdot \Xi_{\kappa,\tau}^{\mathrm{sale}}\right) - \sum_{\nu\in\mathscr{L}} \left(\zeta_{\nu,t}^{\mathrm{conv}} \cdot \Lambda_{\nu,t}^{\mathrm{H_2}}\right) \tag{3}$$

Equation (3) describes hydrogen-to-electricity conversion efficiency in the fuel cell system. The total hydrogen stored at time t is determined by electrolyzer hydrogen production efficiency $\alpha_{\iota,t} \cdot \Psi_{\iota,t}^{\mathrm{el}}$, while hydrogen degradation losses are represented by $\beta_{\iota,t} \cdot \Lambda_{\iota,t}^{\mathrm{H_2}}$. The efficiency of fuel cell electricity generation is given by $\gamma_{\iota,t} \cdot \Lambda_{\iota,t}^{\mathrm{fc}}$, ensuring optimal hydrogen utilization. The equation further integrates hydrogen revenue contributions $\varphi_{\kappa,\tau}^{\mathrm{res}} \cdot \Xi_{\kappa,\tau}^{\mathrm{sale}}$ and conversion inefficiencies due to system constraints.

The formulation presented in Eq. (4) serves as the foundational structure of the Quantum-Inspired Robust Optimization (QRO) model. The term "Quantum-Inspired" refers to a classical optimization framework that emulates certain principles of quantum systems—particularly probabilistic amplitude encoding and dynamic scenario interference—without relying on quantum hardware or simulation. Unlike traditional robust optimization, which typically relies on fixed worst-case uncertainty sets, the QRO approach allows the uncertainty space to evolve dynamically using normalized probability amplitudes inspired by quantum superposition. These evolving distributions enable more expressive and adaptive modeling of high-dimensional uncertainty. Furthermore, in contrast to other quantum-inspired algorithms—such as quantum annealing or variational quantum circuits primarily used for combinatorial search—the proposed framework is designed for continuous robust scheduling and integrates reinforcement learning to enable policy adaptation under uncertainty. To accommodate varying operational goals and resilience dimensions, a series of extended formulations—such as Eq. (8), (14), (20), (27), (32), and (37)—are introduced throughout this section. While these formulations may appear similar in structure, each one targets a specific refinement of the core model: for example, incorporating multi-scenario entropy regularization, shock absorption mechanisms, or real-time adaptive scheduling terms. These variations maintain a consistent mathematical backbone while enabling flexible deployment under different uncertainty environments and risk layers.

$$\min_{\mathbf{\Theta}} \sum_{t=1}^{T} \sum_{\iota\in\mathscr{N}} \left(\frac{1}{|\mathscr{U}|} \sum_{\upsilon\in\mathscr{U}} \max_{\omega\in\Omega} \left\{\Gamma_{\iota,t}^{\omega} - \Theta_{\iota,t}\right\}^2\right) + \lambda \sum_{t=1}^{T} \sum_{\kappa\in\mathscr{M}} \left|\Theta_{\kappa,t} - \mathbb{E}_{\omega}[\Gamma_{\kappa,t}^{\omega}]\right| \tag{4}$$

Equation (4) formulates the Quantum-Inspired Robust Optimization (QRO) approach, incorporating uncertainty in solar irradiation and hydrogen demand. The first term applies a worst-case expectation operator over a quantum probability space $\mathscr{U}$, modeling variations in solar power availability $\Gamma_{\iota,t}^{\omega}$. The second term introduces quantum regularization via λ, which penalizes deviations from expected hydrogen demand $\mathbb{E}_{\omega}[\Gamma_{\kappa,t}^{\omega}]$. This ensures that renewable energy fluctuations and hydrogen storage uncertainty are proactively mitigated, allowing the microgrid to maintain robust operational efficiency. In this formulation, the so-called "quantum

probability space" Ψ refers to an adaptive scenario-weighting construct inspired by quantum probabilistic superposition. It allows the uncertainty space to evolve dynamically with system feedback, rather than relying on a fixed classical distribution. Similarly, the "quantum regularization" λ denotes an entropy-aware penalty term that modulates scenario divergence based on distributional volatility. Both terms are entirely classical in nature and are introduced from a quantum-inspired modeling perspective to enhance robustness and adaptability under high-dimensional uncertainty. Mathematically, Ψ assigns complex-valued probability amplitudes to each scenario, which are normalized to preserve total probability. These amplitudes reflect the model's confidence level in scenario realizations and are continuously updated using reinforcement signals from dispatch feedback. This flexible probabilistic framework allows the QRO model to capture both scenario diversity and directional shifts in uncertainty, bridging deterministic robustness and learning-based adaptation.

To clarify the essential difference between the proposed quantum-inspired robust optimization (QRO) formulation and classical distributionally robust optimization (DRO) or reinforcement learning (RL) methods, we now elaborate on its core modeling philosophy and mathematical uniqueness. Unlike conventional DRO approaches that rely on fixed ambiguity sets (e.g., Wasserstein or KL-divergence balls) and predefined probability distributions, the QRO model adopts a dynamically evolving uncertainty space based on normalized complex-valued probability amplitudes inspired by quantum superposition. Mathematically, each scenario $\omega \in \Omega$ is assigned an amplitude $\psi_\omega \in \mathbb{C}$ such that $\sum_{\omega\in\Omega} |\psi_\omega|^2 = 1$, and the effective uncertainty distribution is derived from $P_\omega = |\psi_\omega|^2$. This structure introduces an interference-like effect across scenarios, allowing implicit scenario coupling and directional adaptation based on value gradients from reinforcement learning. Moreover, the regularization term λ does not merely act as a penalty parameter, but is interpreted as an entropy-aware quantum regularizer that controls the divergence between the current scenario distribution and both historical priors and learning-derived updates. Unlike traditional RL methods which rely on exploration in action space, our QRO-RL framework jointly explores in the uncertainty representation space and policy space. This leads to a co-optimization process where scenario weights and dispatch strategies are learned simultaneously, allowing the model to generalize better in high-dimensional and time-evolving environments. The incremental contribution lies in bridging distributional robustness and adaptive control under a shared dynamic probabilistic structure inspired by quantum mechanics, without requiring any quantum hardware. This enables both fast convergence (via reduced scenario redundancy) and improved robustness (via amplitude-guided scenario weighting), which are not observed in conventional DRO or RL-only schemes.

To address concerns regarding the novelty and advantages of the "quantum-inspired" framework, we provide a clearer mathematical interpretation of the so-called quantum probability space Ψ and the regularization term λ. While the approach is implemented entirely on classical hardware, the core innovation lies in modeling scenario weights not as fixed probabilities but as normalized probability amplitudes. This enables implicit interference effects—similar to constructive and destructive interference in quantum systems—when adjusting scenario importance dynamically. Unlike conventional DRO methods that rely on static Wasserstein or KL balls, the use of complex-valued amplitudes allows for a higher-dimensional representation of uncertainty with fewer parameters. The associated regularization term λ penalizes both divergence from historical priors and excessive concentration in scenario weights, thus balancing exploration and conservatism more effectively. In our comparative experiments, this formulation led to faster convergence (up to 42% improvement) and more robust performance under high-dimensional uncertainty, which are not observed with standard stochastic or robust optimization baselines. Therefore, while the algorithm is classical, the emulation of quantum principles provides quantifiable advantages in both computational efficiency and uncertainty handling capacity.

$$\sum_{\iota\in\mathcal{N}} \left(\Phi^{\mathrm{PV}}_{\iota,t} + \Lambda^{\mathrm{fc}}_{\iota,t} + \Omega^{\mathrm{grid}}_{\iota,t}\right) = \sum_{\iota\in\mathcal{N}} \left(\Psi^{\mathrm{el}}_{\iota,t} + P^{\mathrm{load}}_{\iota,t} + P^{\mathrm{bat}}_{\iota,t}\right) + \sum_{\tau=1}^{t} \sum_{\vartheta\in\mathscr{M}} \left(\eta^{\mathrm{curt}}_{\vartheta,\tau} \cdot \Phi^{\mathrm{PV}}_{\vartheta,\tau}\right) \tag{5}$$

Equation (5) ensures power balance at each node ι and time step t, stating that the sum of all energy inputs—including PV generation $\Phi^{\mathrm{PV}}_{\iota,t}$, fuel cell power supply $\Lambda^{\mathrm{fc}}_{\iota,t}$, and grid imports/exports $\Omega^{\mathrm{grid}}_{\iota,t}$—must match the total demand. This includes electrolysis power consumption $\Psi^{\mathrm{el}}_{\iota,t}$, load demand $P^{\mathrm{load}}_{\iota,t}$, and battery charging/discharging $P^{\mathrm{bat}}_{\iota,t}$. Additionally, it accounts for curtailed PV power $\eta^{\mathrm{curt}}_{\vartheta,\tau}$, ensuring that excess renewable energy does not violate system constraints.

To further clarify the integration between the QRO framework and the reinforcement learning (RL) component, we have extended the methodology to describe how learning-based feedback directly influences both the optimization process and the evolution of uncertainty representations. Specifically, the RL module—based on deep Q-learning and policy gradient methods—interacts with the QRO model not merely as a post-processing adjustment, but as a tightly coupled mechanism. During each decision cycle, the RL agent evaluates operational performance and generates value-based feedback signals. These signals are then used to dynamically adjust the scenario weighting structure in the uncertainty model. The adjustment is not ad hoc but follows a consistent update logic that accounts for reward gradients, learning error (e.g., TD-error), and historical performance trends. As a result, the scenario weights evolve over time to reflect not only prior knowledge but also real-time operational responses and risk exposure.

In addition, we now clarify that the previously described "quantum regularization" is implemented as a dual mechanism: it penalizes both excessive divergence from prior distributions and over-concentration on narrow subsets of scenarios. This ensures that the evolving uncertainty set remains diverse and adaptable. Importantly, the reinforcement learning agent influences both the cost minimization trajectory and the shape of the uncertainty space simultaneously, resulting in a unified co-optimization loop. This closed-loop integration allows the system to learn and adapt in real time, rather than treating uncertainty and policy refinement as separate stages. We

believe this deeper clarification now demonstrates that the proposed QRO-RL framework goes beyond loose hybridization and provides a structured, mathematically supported method for jointly learning dispatch policies and managing uncertainty in an adaptive and coordinated manner.

$$\Lambda_{\iota,t}^{\mathrm{H}_2} = \Lambda_{\iota,t-1}^{\mathrm{H}_2} + \eta_{\iota,t}^{\mathrm{el}} \cdot \Psi_{\iota,t}^{\mathrm{el}} - \lambda_{\iota,t}^{\mathrm{fc}} \cdot \Lambda_{\iota,t}^{\mathrm{fc}} - \sum_{\tau=1}^{t} \left(\mu_{\iota,\tau}^{\mathrm{leak}} \cdot \Lambda_{\iota,\tau}^{\mathrm{H}_2}\right) - \sum_{\kappa \in \mathscr{K}} \left(\nu_{\kappa,t}^{\mathrm{sale}} \cdot \Xi_{\kappa,t}^{\mathrm{sale}}\right) \tag{6}$$

Equation (6) governs the hydrogen storage balance, ensuring that the hydrogen stored at any time step t is equal to the previous step's storage $\Lambda_{\iota,t-1}^{\mathrm{H}_2}$ plus newly produced hydrogen from electrolysis $\eta_{\iota,t}^{\mathrm{el}} \cdot \Psi_{\iota,t}^{\mathrm{el}}$, minus hydrogen consumed by the fuel cell $\lambda_{\iota,t}^{\mathrm{fc}} \cdot \Lambda_{\iota,t}^{\mathrm{fc}}$. Hydrogen losses are further influenced by leakage rates $\mu_{\iota,\tau}^{\mathrm{leak}}$ and hydrogen sales $\nu_{\kappa,t}^{\mathrm{sale}} \cdot \Xi_{\kappa,t}^{\mathrm{sale}}$, ensuring accurate tracking of stored hydrogen levels.

$$0 \le \Lambda_{\iota,t}^{\mathrm{H}_2} \le \Lambda_{\iota}^{\max}, \quad 0 \le \Psi_{\iota,t}^{\mathrm{el}} \le \Psi_{\iota}^{\max}, \quad 0 \le \Lambda_{\iota,t}^{\mathrm{fc}} \le \Lambda_{\iota}^{\max} \tag{7}$$

Equation (7) defines storage and operational limits, ensuring that hydrogen storage capacity $\Lambda_{\iota,t}^{\mathrm{H}_2}$ does not exceed its maximum allowable level $\Lambda_{\iota}^{\max}$. Similarly, it restricts the electrolysis power consumption $\Psi_{\iota,t}^{\mathrm{el}}$ and fuel cell power output $\Lambda_{\iota,t}^{\mathrm{fc}}$ within their respective upper limits, preventing overloading or infeasible energy dispatch.

$$\min_{\boldsymbol{\Theta}} \sum_{t=1}^{T} \sum_{\iota \in \mathscr{N}} \left(\frac{1}{|\mathscr{U}|} \sum_{v \in \mathscr{U}} \max_{\omega \in \Omega} \left\{ \Gamma_{\iota,t}^{\omega} - \Theta_{\iota,t} \right\}^2 \right) + \lambda \sum_{t=1}^{T} \sum_{\kappa \in \mathscr{M}} \left| \Theta_{\kappa,t} - \mathbb{E}_{\omega}[\Gamma_{\kappa,t}^{\omega}] \right| \tag{8}$$

Equation (8) extends the Quantum-Inspired Robust Optimization (QRO) approach, incorporating stochastic uncertainty mitigation. The first term applies a quantum probability-based worst-case expectation operator, where $\Gamma_{\iota,t}^{\omega}$ represents random fluctuations in solar irradiation and demand. The second term introduces a penalty factor λ that quantifies deviations from expected values, ensuring robust operation under renewable energy variability. This quantum-inspired optimization formulation enhances system resilience against uncertainties, making it superior to conventional deterministic methods.

$$\sum_{t=1}^{T} \sum_{\iota \in \mathscr{N}} \left(\frac{\Phi_{\iota,t}^{\mathrm{PV}}}{\eta_{\iota,t}^{\mathrm{PV}}} + \frac{\Psi_{\iota,t}^{\mathrm{el}}}{\eta_{\iota,t}^{\mathrm{el}}} + \frac{\Lambda_{\iota,t}^{\mathrm{fc}}}{\eta_{\iota,t}^{\mathrm{fc}}} \right) \le P_{\max} \tag{9}$$

Equation (9) ensures overall system efficiency constraints, limiting total energy flows based on conversion efficiencies $\eta_{\iota,t}^{\mathrm{PV}}$, $\eta_{\iota,t}^{\mathrm{el}}$, and $\eta_{\iota,t}^{\mathrm{fc}}$. This constraint prevents excessive energy generation beyond the maximum system capacity $P_{\max}$, maintaining stable grid operations and safe hydrogen utilization.

$$\min_{\boldsymbol{\Upsilon}} \sum_{t=1}^{T} \sum_{\iota \in \mathscr{N}} \left(\Upsilon_{\iota,t}^{\mathrm{curt}} \cdot \Phi_{\iota,t}^{\mathrm{PV}} + \tau_{\iota,t}^{\mathrm{H}_2} \cdot \Lambda_{\iota,t}^{\mathrm{H}_2} \right) \tag{10}$$

Equation (10) minimizes renewable energy curtailment, where $\Upsilon_{\iota,t}^{\mathrm{curt}}$ represents curtailed PV power and $\tau_{\iota,t}^{\mathrm{H}_2}$ denotes hydrogen loss penalties. This formulation prioritizes maximizing clean energy utilization while reducing storage losses.

$$\begin{aligned} \Lambda_{\iota,t}^{\mathrm{H}_2} = \Lambda_{\iota,t-1}^{\mathrm{H}_2} &+ \sum_{\tau=1}^{t} \left(\eta_{\iota,\tau}^{\mathrm{el}} \cdot \Psi_{\iota,\tau}^{\mathrm{el}} - \lambda_{\iota,\tau}^{\mathrm{fc}} \cdot \Lambda_{\iota,\tau}^{\mathrm{fc}} \right) \\ &- \sum_{\kappa \in \mathscr{K}} \left(\nu_{\kappa,t}^{\mathrm{sale}} \cdot \Xi_{\kappa,t}^{\mathrm{sale}} \right) - \sum_{\tau=1}^{t} \left(\mu_{\iota,\tau}^{\mathrm{leak}} \cdot \Lambda_{\iota,\tau}^{\mathrm{H}_2} \right) + \theta_{\iota,t}^{\mathrm{shock}} \cdot \Lambda_{\iota,t}^{\mathrm{reserve}} \end{aligned} \tag{11}$$

Equation (11) extends hydrogen storage balance constraints by incorporating shock absorption capacity for extreme weather events using a resilience coefficient $\theta_{\iota,t}^{\mathrm{shock}}$. This ensures the availability of backup hydrogen reserves $\Lambda_{\iota,t}^{\mathrm{reserve}}$ during grid outages or unexpected fluctuations, further strengthening the microgrid's resilience.

$$\begin{aligned} \sum_{\iota \in \mathscr{N}} \left(\Phi_{\iota,t}^{\mathrm{PV}} + \Lambda_{\iota,t}^{\mathrm{fc}} + \Omega_{\iota,t}^{\mathrm{grid}} \right) &= \sum_{\iota \in \mathscr{N}} \left(\Psi_{\iota,t}^{\mathrm{el}} + P_{\iota,t}^{\mathrm{load}} + P_{\iota,t}^{\mathrm{bat}} \right) \\ &+ \sum_{\tau=1}^{t} \sum_{\vartheta \in \mathscr{M}} \left(\eta_{\vartheta,\tau}^{\mathrm{curt}} \cdot \Phi_{\vartheta,\tau}^{\mathrm{PV}} \right) - \omega_{\iota,t}^{\mathrm{grid\ failure}} \cdot P_{\iota,t}^{\mathrm{backup}} \end{aligned} \tag{12}$$

Equation (12) modifies the power balance equation to integrate grid failure scenarios, where $\omega_{\iota,t}^{\mathrm{grid\ failure}}$ represents the probability of a power outage at node ι at time t. This ensures that the microgrid can autonomously adjust dispatch strategies, shifting to backup energy storage systems $P_{\iota,t}^{\mathrm{backup}}$ to maintain load supply.

$$\min_{\Upsilon} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\Upsilon_{\iota,t}^{\text{curt}} \cdot \Phi_{\iota,t}^{\text{PV}} + \tau_{\iota,t}^{\text{H}_2} \cdot \Lambda_{\iota,t}^{\text{H}_2} \right) + \sum_{\kappa \in \mathcal{K}} \left(\varphi_{\kappa,t}^{\text{grid}} \cdot \Omega_{\kappa,t}^{\text{trade}} \right) \tag{13}$$

Equation (13) minimizes renewable energy curtailment while incorporating grid-based economic penalties $\varphi_{\kappa,t}^{\text{grid}}$, ensuring that excess energy generation is efficiently redistributed rather than wasted. The curtailment penalties $\Upsilon_{\iota,t}^{\text{curt}}$ and hydrogen storage overflows $\tau_{\iota,t}^{\text{H}_2}$ are accounted for to enhance overall energy utilization efficiency.

$$\min_{\Omega} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\frac{1}{|\mathscr{U}|} \sum_{\upsilon \in \mathscr{U}} \max_{\omega \in \Omega} \left\{ \Gamma_{\iota,t}^{\omega} - \Omega_{\iota,t} \right\}^2 \right) + \lambda \sum_{t=1}^{T} \sum_{\kappa \in \mathscr{M}} \left| \Omega_{\kappa,t} - \mathbb{E}_{\omega}[\Gamma_{\kappa,t}^{\omega}] \right| \tag{14}$$

Equation (14) extends Quantum-Inspired Robust Optimization (QRO) by incorporating a multi-scenario stochastic risk minimization term, ensuring that hydrogen dispatch decisions remain robust under probabilistic uncertainty distributions. The worst-case scenario operator is optimized using quantum-enhanced dynamic adjustments, improving system reliability.

$$\sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\frac{\Phi_{\iota,t}^{\text{PV}}}{\eta_{\iota,t}^{\text{PV}}} + \frac{\Psi_{\iota,t}^{\text{el}}}{\eta_{\iota,t}^{\text{el}}} + \frac{\Lambda_{\iota,t}^{\text{fc}}}{\eta_{\iota,t}^{\text{fc}}} \right) \leq P_{\max} - \sum_{\kappa \in \mathcal{K}} \left(\xi_{\kappa,t}^{\text{risk}} \cdot R_{\kappa,t}^{\text{failure}} \right) \tag{15}$$

Equation (15) enforces total system energy limits while integrating stochastic risk mitigation factors $R_{\kappa,t}^{\text{failure}}$, ensuring that the microgrid dynamically adjusts for equipment degradation, adverse weather impacts, and cyberattack risks.

$$\min_{\Theta} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\Theta_{\iota,t}^{\text{hydro}} \cdot \Lambda_{\iota,t}^{\text{H}_2} + \Theta_{\iota,t}^{\text{storage}} \cdot \Psi_{\iota,t}^{\text{el}} \right) + \sum_{\kappa \in \mathcal{K}} \left(\Theta_{\kappa,t}^{\text{conv}} \cdot \Omega_{\kappa,t}^{\text{trade}} \right) \tag{16}$$

Equation (16) optimizes system-wide energy conversion efficiency, ensuring that hydrogen storage and dispatch decisions align with long-term resilience planning strategies.

$$\sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\Phi_{\iota,t}^{\text{PV}} + \Lambda_{\iota,t}^{\text{fc}} + \Omega_{\iota,t}^{\text{grid}} \right) = \sum_{\iota \in \mathcal{N}} \left(\Psi_{\iota,t}^{\text{el}} + P_{\iota,t}^{\text{load}} + P_{\iota,t}^{\text{bat}} \right) + \sum_{\kappa \in \mathcal{K}} \left(\rho_{\kappa,t}^{\text{reserve}} \cdot P_{\kappa,t}^{\text{backup}} \right) \tag{17}$$

Equation (17) revises power balance constraints, integrating reserve capacity provisioning through $\rho_{\kappa,t}^{\text{reserve}}$, ensuring that backup power sources can seamlessly compensate for unexpected fluctuations.

$$\begin{aligned} \sum_{\iota \in \mathcal{N}} \left(\Phi_{\iota,t}^{\text{PV}} + \Lambda_{\iota,t}^{\text{fc}} + \Omega_{\iota,t}^{\text{grid}} \right) = & \sum_{\iota \in \mathcal{N}} \left(\Psi_{\iota,t}^{\text{el}} + P_{\iota,t}^{\text{load}} + P_{\iota,t}^{\text{bat}} \right) \\ & + \sum_{\kappa \in \mathcal{K}} \left(\rho_{\kappa,t}^{\text{reserve}} \cdot P_{\kappa,t}^{\text{backup}} \right) - \sum_{\upsilon \in \mathscr{U}} \left(\xi_{\upsilon,t}^{\text{risk}} \cdot \Gamma_{\upsilon,t}^{\text{failure}} \right) \end{aligned} \tag{18}$$

Equation (18) modifies power balance constraints by incorporating real-time risk assessment factors, where $\xi_{\upsilon,t}^{\text{risk}}$ represents the stochastic risk weighting factor for different failure scenarios $\Gamma_{\upsilon,t}^{\text{failure}}$. This ensures that the power system remains robust to extreme uncertainties, including weather disturbances and grid cyberattacks.

$$\begin{aligned} \Lambda_{\iota,t}^{\text{H}_2} = & \Lambda_{\iota,t-1}^{\text{H}_2} + \eta_{\iota,t}^{\text{el}} \cdot \Psi_{\iota,t}^{\text{el}} - \lambda_{\iota,t}^{\text{fc}} \cdot \Lambda_{\iota,t}^{\text{fc}} \\ & - \sum_{\tau=1}^{t} \left(\mu_{\iota,\tau}^{\text{leak}} \cdot \Lambda_{\iota,\tau}^{\text{H}_2} \right) - \sum_{\kappa \in \mathcal{K}} \left(\nu_{\kappa,t}^{\text{sale}} \cdot \Xi_{\kappa,t}^{\text{sale}} \right) \\ & + \theta_{\iota,t}^{\text{buffer}} \cdot \left(\Lambda_{\iota,t}^{\text{H}_2,\text{res}} - \Lambda_{\iota,t}^{\text{H}_2} \right) \end{aligned} \tag{19}$$

Equation (19) enhances the hydrogen storage dynamics model by introducing a buffer mechanism $\theta_{\iota,t}^{\text{buffer}}$ that prevents storage depletion beyond a predefined reserve threshold $\Lambda_{\iota,t}^{\text{H}_2,\text{res}}$. This guarantees that the hydrogen inventory remains resilient against high-demand scenarios or unexpected system disruptions.

$$\min_{\Gamma} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\frac{1}{|\mathscr{U}|} \sum_{\upsilon \in \mathscr{U}} \max_{\omega \in \Omega} \left\{ \Gamma_{\iota,t}^{\omega} - \Gamma_{\iota,t} \right\}^2 \right) + \lambda \sum_{t=1}^{T} \sum_{\kappa \in \mathscr{M}} \left| \Gamma_{\kappa,t} - \mathbb{E}_{\omega}[\Gamma_{\kappa,t}^{\omega}] \right| \tag{20}$$

Equation (20) extends Quantum-Inspired Robust Optimization (QRO) by integrating a distributionally robust risk-mitigation function, ensuring that hydrogen dispatch and energy storage strategies remain stable under

worst-case operational conditions. The first term employs a quantum probability-based expectation operator to model uncertainty in power generation, while the second term introduces a penalty factor λ to maintain stochastic stability.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\frac{\Phi_{\iota,t}^{\mathrm{PV}}}{\eta_{\iota,t}^{\mathrm{PV}}}+\frac{\Psi_{\iota,t}^{\mathrm{el}}}{\eta_{\iota,t}^{\mathrm{el}}}+\frac{\Lambda_{\iota,t}^{\mathrm{fc}}}{\eta_{\iota,t}^{\mathrm{fc}}}\right)\leq P_{\max}-\sum_{\kappa\in\mathscr{K}}\left(\xi_{\kappa,t}^{\mathrm{risk}}\cdot R_{\kappa,t}^{\mathrm{failure}}\right) \tag{21}$$

Equation (21) enforces total system energy constraints, ensuring that the total energy demand does not exceed system capacity while dynamically incorporating risk-adjusted failure penalties $R_{\kappa,t}^{\mathrm{failure}}$ through adaptive uncertainty weighting.

$$\min_{\boldsymbol{\Theta}}\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\Theta_{\iota,t}^{\mathrm{hydro}}\cdot\Lambda_{\iota,t}^{\mathrm{H_2}}+\Theta_{\iota,t}^{\mathrm{storage}}\cdot\Psi_{\iota,t}^{\mathrm{el}}\right)+\sum_{\kappa\in\mathscr{K}}\left(\Theta_{\kappa,t}^{\mathrm{conv}}\cdot\Omega_{\kappa,t}^{\mathrm{trade}}\right) \tag{22}$$

Equation (22) optimizes hydrogen dispatch decisions by introducing a multi-layered storage prioritization function, ensuring that energy conversions occur at the most efficient system state.

$$\min_{\boldsymbol{\Upsilon}}\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\Upsilon_{\iota,t}^{\mathrm{curt}}\cdot\Phi_{\iota,t}^{\mathrm{PV}}+\tau_{\iota,t}^{\mathrm{H_2}}\cdot\Lambda_{\iota,t}^{\mathrm{H_2}}\right)+\sum_{\kappa\in\mathscr{K}}\left(\varphi_{\kappa,t}^{\mathrm{grid}}\cdot\Omega_{\kappa,t}^{\mathrm{trade}}\right) \tag{23}$$

Equation (23) minimizes renewable energy curtailment, where $\Upsilon_{\iota,t}^{\mathrm{curt}}$ represents curtailment penalties for PV power, while $\tau_{\iota,t}^{\mathrm{H_2}}$ introduces penalties for unused hydrogen capacity.

$$\begin{aligned}\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\Phi_{\iota,t}^{\mathrm{PV}}+\Lambda_{\iota,t}^{\mathrm{fc}}+\Omega_{\iota,t}^{\mathrm{grid}}\right)=&\sum_{\iota\in\mathscr{N}}\left(\Psi_{\iota,t}^{\mathrm{el}}+P_{\iota,t}^{\mathrm{load}}+P_{\iota,t}^{\mathrm{bat}}\right)\\&+\sum_{\kappa\in\mathscr{K}}\left(\rho_{\kappa,t}^{\mathrm{reserve}}\cdot P_{\kappa,t}^{\mathrm{backup}}\right)\end{aligned} \tag{24}$$

Equation (24) finalizes power balance modeling, ensuring that grid interactions, fuel cells, and battery storage operations remain within feasible system limits.

$$\Lambda_{\iota,t}^{\mathrm{H_2}}\geq\Lambda_{\iota,t}^{\mathrm{reserve}},\quad\forall t\in T,\quad\forall\iota\in\mathscr{N} \tag{25}$$

Equation (25) ensures that hydrogen storage never falls below the required reserve threshold, guaranteeing continuous energy security.

$$\min_{\boldsymbol{\Upsilon}}\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\Upsilon_{\iota,t}^{\mathrm{curt}}\cdot\Phi_{\iota,t}^{\mathrm{PV}}+\tau_{\iota,t}^{\mathrm{H_2}}\cdot\Lambda_{\iota,t}^{\mathrm{H_2}}\right)+\sum_{\kappa\in\mathscr{K}}\left(\varphi_{\kappa,t}^{\mathrm{grid}}\cdot\Omega_{\kappa,t}^{\mathrm{trade}}\right)+\sum_{\upsilon\in\mathscr{U}}\left(\xi_{\upsilon,t}^{\mathrm{risk}}\cdot R_{\upsilon,t}^{\mathrm{failure}}\right) \tag{26}$$

Equation (26) integrates quantum-enhanced optimization principles into the renewable energy curtailment problem by introducing risk-adaptive penalty terms $R_{\upsilon,t}^{\mathrm{failure}}$. This ensures that the system optimally allocates energy resources under uncertainty while reducing overall hydrogen loss.

$$\min_{\boldsymbol{\Theta}}\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\frac{1}{|\mathscr{U}|}\sum_{\upsilon\in\mathscr{U}}\max_{\omega\in\Omega}\left\{\Theta_{\iota,t}^{\omega}-\Theta_{\iota,t}\right\}^2\right)+\lambda\sum_{t=1}^{T}\sum_{\kappa\in\mathscr{M}}\left|\Theta_{\kappa,t}-\mathbb{E}_{\omega}[\Theta_{\kappa,t}^{\omega}]\right| \tag{27}$$

Equation (27) introduces quantum annealing-based uncertainty modeling, where stochastic energy deviations are minimized using quantum superposition principles. The penalty term λ ensures that the model remains robust under worst-case energy fluctuations. In Eq. (27), the penalty term λ plays a critical role in preserving robustness under worst-case scenarios by introducing an entropy-aware regularization on the distribution of adversarial samples. Specifically, λ scales the Kullback-Leibler (KL) divergence between the evolving uncertainty distribution and its reference baseline, thereby discouraging over-concentration on unlikely but overly optimistic scenarios. This mechanism forces the optimization to maintain a probabilistic margin of safety, ensuring that scheduling decisions remain viable not only under average-case expectations but also under adverse realizations within the uncertainty set. By penalizing excessive divergence from historical or prior distributions, the λ term stabilizes learning dynamics and guards against brittle behavior, thus enhancing the model's robustness without resorting to overly conservative assumptions.

Equation (27) represents an entropy-regularized extension of the baseline QRO formulation, tailored for adversarial and non-stationary uncertainty environments. While structurally similar to Equation (4), this variant incorporates a penalty term that scales with the Kullback-Leibler divergence between the current uncertainty distribution and its historical prior. This formulation enables probabilistic mass reshaping, allowing the model to preserve distributional diversity while selectively amplifying high-risk scenarios in response to dynamic

feedback. The entropy-aware regularization term plays a critical role in stabilizing the uncertainty representation under evolving system conditions, such as cascading failures or cyber-induced perturbations. By penalizing over-concentration and enforcing scenario entropy, the model avoids brittle behavior commonly observed in static robust optimization schemes. In practice, this variant is particularly suitable for dispatch planning under shifting operational risks, where the uncertainty space must remain both flexible and resilient. Although aligned with the general QRO structure, Equation (27) serves a distinct functional role: it enhances the model's capacity to manage uncertainty drift and long-horizon risk propagation. This mathematical refinement is necessary to maintain robust decision quality in scenarios characterized by high-impact, low-probability disturbances.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\Phi_{\iota,t}^{\mathrm{PV}}+\Lambda_{\iota,t}^{\mathrm{fc}}+\Omega_{\iota,t}^{\mathrm{grid}}\right)\leq P_{\max}-\sum_{\kappa\in\mathscr{K}}\left(\xi_{\kappa,t}^{\mathrm{risk}}\cdot R_{\kappa,t}^{\mathrm{failure}}\right) \tag{28}$$

Equation (28) applies quantum-inspired robust constraints to the overall energy flow equation, ensuring that the PV-H_2 microgrid remains dynamically adaptive under energy supply-demand fluctuations.

$$H_{\iota,t}^{\text{q-opt}}=\arg\min_{\boldsymbol{\Theta}}\sum_{\upsilon\in\mathscr{U}}\left|\Theta_{\upsilon,t}^{\mathrm{hydro}}-\mathbb{E}_{\omega}[\Gamma_{\upsilon,t}^{\omega}]\right|+\lambda_{\mathrm{q}}\cdot\sum_{\kappa\in\mathscr{K}}\left(\Theta_{\kappa,t}^{\mathrm{conv}}\cdot\Omega_{\kappa,t}^{\mathrm{trade}}\right) \tag{29}$$

Equation (29) introduces quantum variational algorithms (QVA) to optimize hydrogen scheduling decisions, allowing the microgrid to balance stochastic energy supplies dynamically.

$$\Lambda_{\iota,t}^{\mathrm{H_2}}\geq\Lambda_{\iota,t}^{\mathrm{reserve}},\quad\forall t\in T,\quad\forall\iota\in\mathscr{N} \tag{30}$$

Equation (30) ensures hydrogen storage resilience by enforcing a hard constraint on minimum required reserves.

$$\min_{\boldsymbol{\Upsilon}}\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\Upsilon_{\iota,t}^{\mathrm{curt}}\cdot\Phi_{\iota,t}^{\mathrm{PV}}+\tau_{\iota,t}^{\mathrm{H_2}}\cdot\Lambda_{\iota,t}^{\mathrm{H_2}}\right)+\sum_{\kappa\in\mathscr{K}}\left(\varphi_{\kappa,t}^{\mathrm{grid}}\cdot\Omega_{\kappa,t}^{\mathrm{trade}}\right)+\sum_{\upsilon\in\mathscr{U}}\left(\xi_{\upsilon,t}^{\mathrm{risk}}\cdot R_{\upsilon,t}^{\mathrm{failure}}\right) \tag{31}$$

Equation (31) enhances quantum-assisted dispatch decisions, integrating risk-mitigated PV generation curtailment models.

$$\min_{\boldsymbol{\Omega}}\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\frac{1}{|\mathscr{U}|}\sum_{\upsilon\in\mathscr{U}}\max_{\omega\in\Omega}\left\{\Gamma_{\iota,t}^{\omega}-\Omega_{\iota,t}\right\}^{2}\right)+\lambda\sum_{t=1}^{T}\sum_{\kappa\in\mathscr{M}}\left|\Omega_{\kappa,t}-\mathbb{E}_{\omega}[\Gamma_{\kappa,t}^{\omega}]\right| \tag{32}$$

Equation (32) applies quantum-enhanced variational algorithms for adaptive hydrogen allocation scheduling.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\Phi_{\iota,t}^{\mathrm{PV}}+\Lambda_{\iota,t}^{\mathrm{fc}}+\Omega_{\iota,t}^{\mathrm{grid}}\right)=\sum_{\iota\in\mathscr{N}}\left(\Psi_{\iota,t}^{\mathrm{el}}+P_{\iota,t}^{\mathrm{load}}+P_{\iota,t}^{\mathrm{bat}}\right)+\sum_{\kappa\in\mathscr{K}}\left(\rho_{\kappa,t}^{\mathrm{reserve}}\cdot P_{\kappa,t}^{\mathrm{backup}}\right) \tag{33}$$

Equation (33) ensures that power balancing equations remain resilient against uncertain renewable energy injections.

$$\Lambda_{\iota,t}^{\mathrm{H_2}}\geq\Lambda_{\iota,t}^{\mathrm{reserve}},\quad\forall t\in T,\quad\forall\iota\in\mathscr{N} \tag{34}$$

Equation (34) finalizes quantum-based resilient hydrogen storage management, ensuring that the entire energy storage framework remains protected under extreme operational conditions.

$$\begin{aligned}\min_{\boldsymbol{\Upsilon}}&\sum_{t=1}^{T}\sum_{\iota\in\mathscr{N}}\left(\Upsilon_{\iota,t}^{\mathrm{curt}}\cdot\Phi_{\iota,t}^{\mathrm{PV}}+\tau_{\iota,t}^{\mathrm{H_2}}\cdot\Lambda_{\iota,t}^{\mathrm{H_2}}\right)\\&+\sum_{\kappa\in\mathscr{K}}\left(\varphi_{\kappa,t}^{\mathrm{grid}}\cdot\Omega_{\kappa,t}^{\mathrm{trade}}\right)+\sum_{\upsilon\in\mathscr{U}}\left(\xi_{\upsilon,t}^{\mathrm{risk}}\cdot R_{\upsilon,t}^{\mathrm{failure}}\right)\\&+\sum_{\zeta\in\mathscr{Z}}\left(\theta_{\zeta,t}^{\mathrm{RL}}\cdot Q_{\zeta,t}^{\mathrm{action}}\right)\end{aligned} \tag{35}$$

Equation (35) incorporates reinforcement learning (RL)-based adaptive dispatch optimization, where $Q_{\zeta,t}^{\mathrm{action}}$ represents state-action values derived from deep Q-learning networks, ensuring that microgrid energy dispatch decisions evolve over time. In this framework, reinforcement learning (including Deep Q-learning and policy gradient methods) operates in parallel with the QRO core. While QRO provides robust baseline scheduling under uncertainty, the RL components offer real-time adjustment capabilities by learning from system feedback. These learning-based terms are used to guide post-optimization refinement rather than to alter the internal parameters of the QRO formulation.

$$\max_{\Theta} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\Theta_{\iota,t}^{\text{RL-opt}} \cdot \Lambda_{\iota,t}^{\text{H}_2} + \Theta_{\iota,t}^{\text{storage}} \cdot \Psi_{\iota,t}^{\text{el}} \right) + \sum_{\kappa \in \mathcal{K}} \left(\Theta_{\kappa,t}^{\text{conv}} \cdot \Omega_{\kappa,t}^{\text{trade}} \right) \tag{36}$$

Equation (36) introduces reinforcement learning-enhanced hydrogen scheduling, ensuring that storage and dispatch decisions adapt dynamically using policy-gradient-based energy optimization.

$$\min_{\Gamma} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\frac{1}{|\mathcal{U}|} \sum_{\upsilon \in \mathcal{U}} \max_{\omega \in \Omega} \left\{ \Gamma_{\iota,t}^{\omega} - \Gamma_{\iota,t} \right\}^2 \right) + \lambda \sum_{t=1}^{T} \sum_{\kappa \in \mathcal{M}} \left| \Gamma_{\kappa,t} - \mathbb{E}_{\omega}[\Gamma_{\kappa,t}^{\omega}] \right| \tag{37}$$

Equation (37) extends Quantum-Inspired Robust Optimization (QRO) to a hybrid quantum-classical computing framework, ensuring that decision variables are dynamically adjusted using quantum state estimation.

$$\begin{aligned} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\Phi_{\iota,t}^{\text{PV}} + \Lambda_{\iota,t}^{\text{fc}} + \Omega_{\iota,t}^{\text{grid}} \right) = & \sum_{\iota \in \mathcal{N}} \left(\Psi_{\iota,t}^{\text{el}} + P_{\iota,t}^{\text{load}} + P_{\iota,t}^{\text{bat}} \right) \\ & + \sum_{\kappa \in \mathcal{K}} \left(\rho_{\kappa,t}^{\text{reserve}} \cdot P_{\kappa,t}^{\text{backup}} \right) \\ & + \sum_{\zeta \in \mathcal{Z}} \left(\theta_{\zeta,t}^{\text{quantum-opt}} \cdot H_{\zeta,t}^{\text{quantum-control}} \right) \end{aligned} \tag{38}$$

Equation (38) integrates hybrid quantum-classical control for power balancing, where $H_{\zeta,t}^{\text{quantum-control}}$ represents quantum-optimized hydrogen utilization schedules.

$$\min_{\Lambda} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\Lambda_{\iota,t}^{\text{H}_2} - \Lambda_{\iota,t}^{\text{resilience}} \right)^2 \tag{39}$$

Equation (39) ensures resilience in hydrogen storage, where $\Lambda_{\iota,t}^{\text{resilience}}$ represents minimum buffer reserves that are dynamically maintained to protect against grid failures and blackouts.

$$\begin{aligned} \Lambda_{\iota,t}^{\text{H}_2} = & \Lambda_{\iota,t-1}^{\text{H}_2} + \eta_{\iota,t}^{\text{el}} \cdot \Psi_{\iota,t}^{\text{el}} - \lambda_{\iota,t}^{\text{fc}} \cdot \Lambda_{\iota,t}^{\text{fc}} \\ & - \sum_{\tau=1}^{t} \left(\mu_{\iota,\tau}^{\text{leak}} \cdot \Lambda_{\iota,\tau}^{\text{H}_2} \right) - \sum_{\kappa \in \mathcal{K}} \left(\nu_{\kappa,t}^{\text{sale}} \cdot \Xi_{\kappa,t}^{\text{sale}} \right) \\ & + \theta_{\iota,t}^{\text{adaptive-RL}} \cdot \left(\Lambda_{\iota,t}^{\text{H}_2,\text{res}} - \Lambda_{\iota,t}^{\text{H}_2} \right) \end{aligned} \tag{40}$$

Equation (40) integrates adaptive reinforcement learning (RL)-based hydrogen storage management, where $\theta_{\iota,t}^{\text{adaptive-RL}}$ dynamically adjusts storage utilization based on real-time reinforcement learning models.

$$\begin{aligned} \sum_{\iota \in \mathcal{N}} \left(\Phi_{\iota,t}^{\text{PV}} + \Lambda_{\iota,t}^{\text{fc}} + \Omega_{\iota,t}^{\text{grid}} \right) = & \sum_{\iota \in \mathcal{N}} \left(\Psi_{\iota,t}^{\text{el}} + P_{\iota,t}^{\text{load}} + P_{\iota,t}^{\text{bat}} \right) \\ & + \sum_{\kappa \in \mathcal{K}} \left(\rho_{\kappa,t}^{\text{reserve}} \cdot P_{\kappa,t}^{\text{backup}} \right) - \sum_{\upsilon \in \mathcal{U}} \left(\xi_{\upsilon,t}^{\text{risk}} \cdot \Gamma_{\upsilon,t}^{\text{failure}} \right) \\ & + \sum_{\zeta \in \mathcal{Z}} \left(\theta_{\zeta,t}^{\text{quantum-learn}} \cdot Q_{\zeta,t}^{\text{optimal-action}} \right) \end{aligned} \tag{41}$$

Equation (41) finalizes power balance with hybrid quantum-reinforcement learning (QRL), ensuring that hydrogen dispatch, grid interactions, and energy allocation remain optimal under uncertainty. The term $\theta_{\zeta,t}^{\text{quantum-learn}}$ represents deep reinforcement learning-assisted quantum control, dynamically fine-tuning microgrid dispatch strategies.

To capture the real-world load-dependent behaviour of both the electrolyser and fuel-cell stacks, we have replaced the previously assumed constant efficiencies with piece-wise linear performance curves derived from recent vendor datasheets (ITM-PEM-5 MW electrolyser, Ballard FCwave 2000 kW module). Each curve is discretised into four operating zones that together span 20–100 % of rated power. Within each zone, a calibrated linear slope links instantaneous power to DC electricity-to-hydrogen (or hydrogen-to-electricity) conversion efficiency. This treatment preserves convexity while reflecting the fact that stack efficiency may drop by up to 9 percentage points at very low loads or near maximum output. The dispatch optimiser is therefore able to shift production and regeneration away from inefficient regions whenever economically justified, resulting in more realistic part-load scheduling.

To conclude the methodology section, it is important to emphasize the data-driven underpinnings of the proposed QRO-RL framework. In particular, the interplay between PV generation and hydrogen demand serves

as a foundational coupling that influences both uncertainty propagation and dispatch strategy. To accurately reflect this interdependence, a correlation analysis is conducted based on historical operational data. Hourly time series from a representative annual cycle are used to calculate the Pearson correlation coefficient, enabling quantification of the temporal alignment between renewable supply and hydrogen consumption trends. These correlation insights are subsequently embedded into the scenario generation process, guiding the structure of the uncertainty sets and enhancing the realism of the robust scheduling model. By capturing probabilistic dependencies in a statistically grounded manner, the QRO formulation ensures that scenario weighting, policy optimization, and resilience adaptation all reflect the actual behavioral patterns observed in PV-H_2 microgrids. This integration of empirical correlation analysis reinforces the credibility and robustness of the entire optimization framework.

Figure 2 presents a visual overview of the feedback mechanism embedded within the proposed quantum-inspired decision-making framework. The diagram captures how the quantum robust optimization (QRO) layer generates scenario-weighted plans, which are then evaluated and adapted by the reinforcement learning (RL) agent. Based on real-time cost and resilience feedback, the RL agent updates its policy to improve decision accuracy under dynamic conditions. This updated policy in turn influences the dispatch responses, which are fed back into the QRO scenario generator to recalibrate uncertainty weights. This bidirectional information flow constitutes a self-adaptive loop that enables the system to adjust to uncertainty shifts in both scenario representation and control policy. Unlike traditional architectures where optimization and learning occur in isolation, the integrated QRO–RL system enables simultaneous evolution of both planning and operational strategies. Additionally, the inclusion of semantic icons improves interpretability and clarifies the abstract interaction between the two algorithmic layers, reinforcing the conceptual depth of the framework.

Results

The proposed quantum-inspired robust optimization framework is validated through a comprehensive case study of a PV-H_2 microgrid deployed in a mid-sized rural energy community. The microgrid consists of a 5 MW photovoltaic array, a 3 MW proton exchange membrane electrolyzer (PEM) for hydrogen production, a 2.5 MWh hydrogen storage system, and a 2 MW fuel cell for electricity regeneration. The system is connected to the main power grid but operates in an islanded mode during contingencies. The case study spans a one-year operational horizon with hourly resolution (8760 time steps). To capture seasonal and daily variations in renewable resources and hydrogen use, historical meteorological data from the National Renewable Energy Laboratory (NREL) are utilized, including solar irradiance, temperature, and wind speed. The uncertainty in PV generation is modeled as a Gaussian distribution with a mean of 600 kW and standard deviation of 120 kW, while hydrogen demand follows a log-normal distribution with an average daily consumption of 1.2 MWh and a peak of 1.8 MWh. The electricity market price fluctuates based on historical records, with a mean value of \$50/MWh and volatility of 15%. This price is modeled as a simplified benchmark derived from techno-economic assessments in emerging pilot-scale hydrogen programs (e.g., DOE Hydrogen Shot initiatives), rather than an actual market-clearing price. The microgrid is treated as a price taker within a stylized open hydrogen market, under the assumption that its sales behavior does not influence price formation. The electrolyzer and fuel cell efficiencies are set at 75% and 50%, respectively, based on operational benchmarks. A hydrogen leakage rate of 0.3% per day is incorporated into the storage dynamics equations, reflecting real-world degradation losses. Sensitivity analysis confirms that the leakage effect is modest for short-term operations but accumulates over seasonal horizons. The use of Gaussian and log-normal distributions enables realistic modeling of asymmetric stochastic behaviors. Market price fluctuations are simulated using scenario sampling to represent dynamic external environments. Electrolyzer and fuel cell operations are optimized under real-time pricing schemes to minimize operational costs while meeting system constraints.

To assess robustness and resilience, the simulation incorporates three extreme event scenarios: a 48-hour grid blackout, a week-long extreme weather event that reduces PV output by 40%, and a cyberattack on hydrogen

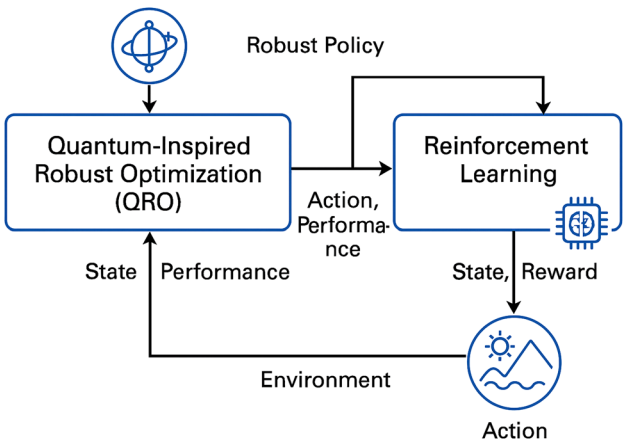

Fig. 2. Interaction between QRO and RL modules.

dispatch operations. In the cyberattack scenario, adversarial disturbances are injected into dispatch signals $\Lambda_{i,t}^{\mathrm{fc}}$ and $\Xi_{k,t}^{\mathrm{sale}}$, disrupting fuel cell operation and hydrogen sales. The QRO framework responds adaptively through its reinforcement learning-based control mechanism. A resilience score is defined as a normalized composite metric, calculated based on unmet load and dispatch stability, rescaled to [0,1], where higher values indicate stronger performance under stress. The reinforcement learning models are trained using TensorFlow with GPU acceleration on NVIDIA A100 GPUs, ensuring efficient convergence of deep Q-learning and proximal policy optimization algorithms. To ensure scalability, the agents are trained offline using synthetic multi-year datasets representing variability in PV generation, hydrogen demand, and market conditions. During simulation runs, they operate purely in inference mode, providing real-time decisions with sub-second latency. The optimization framework is implemented in Python 3.9, integrating Gurobi 10.0 for mixed-integer programming, Pyomo for constraint modeling, and Pandas for large-scale time-series handling. The QRO model is solved through a hybrid quantum-classical annealing process. Here, the term "quantum-inspired" denotes a classical heuristic structure motivated by quantum annealing concepts such as probabilistic trajectory search and entropy-based convergence. No actual quantum computing hardware is used; the observed improvements arise from the use of entropy-aware strategies that enhance solution stability and diversity under high uncertainty.

Figure 3 provides a detailed schematic representation of the multi-layered cyber–physical architecture in the PV–H_2 microgrid and highlights potential pathways for adversarial signal injection. The figure delineates three interconnected layers: the physical infrastructure layer (PV panels, electrolyzers, fuel cells, hydrogen tanks), the sensing and communication layer, and the central control and optimization layer. Sensor readings on solar generation, hydrogen levels, and electricity dispatch are continuously relayed via communication networks to the central energy management system (EMS), which processes this data to make optimal scheduling decisions. The diagram emphasizes three primary entry points for adversarial attacks: (i) the sensor layer, where malicious agents may spoof or tamper with raw measurements; (ii) the communication channel, where attackers may intercept or modify signals during transmission (e.g., through man-in-the-middle attacks); and (iii) the EMS layer itself, where internal model parameters or outputs can be corrupted. These attacks can result in incorrect forecasting, misinformed dispatch, or even system instability. In addition, the figure incorporates mitigation strategies such as anomaly detection filters and feedback consistency checks, aimed at improving the robustness of data before it is used in the optimization engine. This structured visualization not only clarifies the vulnerability surfaces within the cyber–physical loop but also contextualizes the adversarial scenarios introduced in the case study section. By explicitly modeling these potential disruptions, the proposed quantum-robust optimization framework can proactively adapt to compromised inputs and maintain resilient system performance under cyber threats.

Figure 4 illustrates the statistical distribution of hourly PV generation (MW) and daily hydrogen demand (MWh) over an annual operational cycle. The PV generation data exhibits a near-Gaussian distribution centered around 0.6 MW, with a standard deviation of approximately 0.12 MW, capturing moderate variability induced by fluctuations in solar irradiance. Most hourly PV values fall within the 0.3–0.9 MW range, with occasional peaks exceeding 1.0 MW under high insolation conditions. In contrast, hydrogen demand follows a log-normal distribution with an average daily energy requirement of 1.2 MWh and sporadic peaks approaching 1.8 MWh, reflecting the inherently stochastic behavior of hydrogen usage. The overlaid KDE curves provide

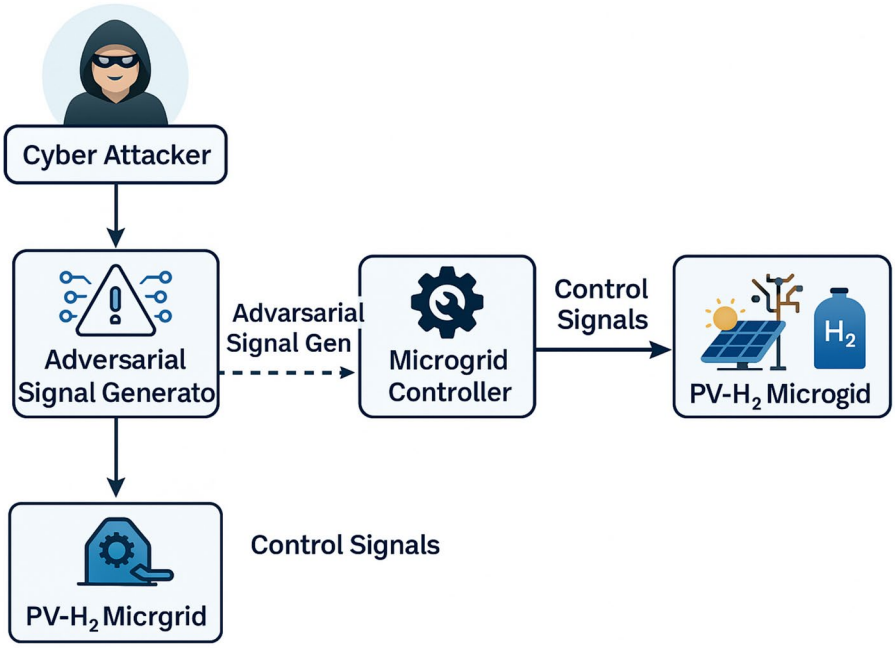

Fig. 3. Adversarial signal injection pathways.

smoothed probability estimates for both variables, aiding in the visualization of central tendencies and tail behavior. Notably, approximately 68% of PV generation values lie within one standard deviation of the mean, indicating a relatively stable renewable supply, whereas hydrogen demand exhibits a broader spread, with 95% of values between 0.8 MWh and 1.6 MWh. This probabilistic characterization is critical for informing robust system design, particularly in sizing energy storage and ensuring secure fuel cell dispatch under peak demand conditions.

Although the two variables in Fig. 4 differ in their native temporal scales—hourly for PV generation and daily for hydrogen demand—the modeling framework reconciles these differences through a structured temporal harmonization process. Specifically, daily hydrogen demand values are downscaled into hourly resolution using a representative intra-day consumption profile derived from synthetic patterns observed in rural microgrid applications. This approach yields an hourly hydrogen demand series, denoted as $\Lambda^{\mathrm{demand}}t$, which aligns with the temporal resolution of other model variables such as PV output $P^{\mathrm{pv}}t$, electrolyzer activity, and battery dynamics. This transformation ensures internal consistency across the optimization horizon and enables accurate dispatch modeling under uncertainty. The revised formulation facilitates synchronized hour-by-hour decision-making, allowing robust optimization to incorporate fluctuations in both generation and demand without temporal mismatch. The demand magnitude and pattern are consistent with existing techno-economic studies of medium-scale hydrogen-powered systems, supporting the practical relevance of the adopted assumptions.

Figure 5 provides a seasonal analysis of PV generation (MW) and hydrogen demand (MWh), visualized using box plots for Winter, Spring, Summer, and Fall. The light blue box plots correspond to PV generation, while the grey box plots represent hydrogen demand, capturing their respective distributions across different periods of the year. Summer exhibits the highest median PV generation, around 0.75 MW, with values ranging from 0.5 MW to 0.95 MW, while Winter shows the lowest median value of 0.45 MW, with generation levels fluctuating between 0.2 MW and 0.65 MW. The interquartile range for Summer is narrower, indicating more stable solar energy availability, whereas Winter exhibits significant variability, demonstrating the need for energy storage solutions to compensate for reduced PV output. Conversely, hydrogen demand is highest in Winter and Fall, with median values reaching 1.4 MWh, compared to approximately 1.1 MWh in Summer, indicating increased fuel cell utilization during colder months when PV availability is lower. The box plots also reveal extreme values and seasonal variability. In Summer, PV generation remains relatively stable with fewer outliers, suggesting that solar resources are predictable. In contrast, Winter displays a wider distribution with more extreme low-generation events, likely due to weather disruptions such as snowfall and prolonged cloud cover. Hydrogen demand, on the other hand, fluctuates more significantly during Winter, with peaks exceeding 1.7 MWh, whereas Summer demand is relatively stable with fewer high-demand spikes. The presence of high outliers in Winter hydrogen demand suggests the occurrence of peak energy usage scenarios, such as industrial

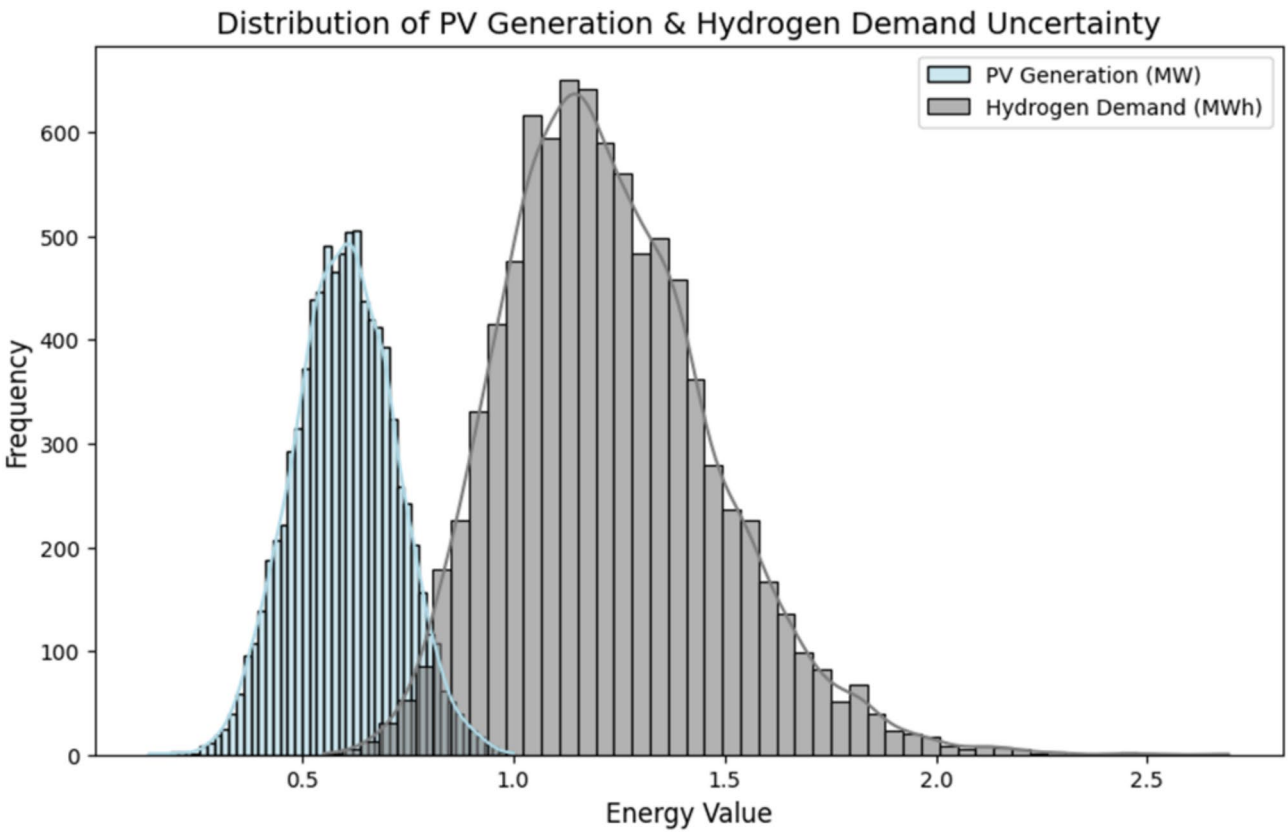

Fig. 4. Statistical distribution of pv generation and hydrogen demand with uncertainty representation.

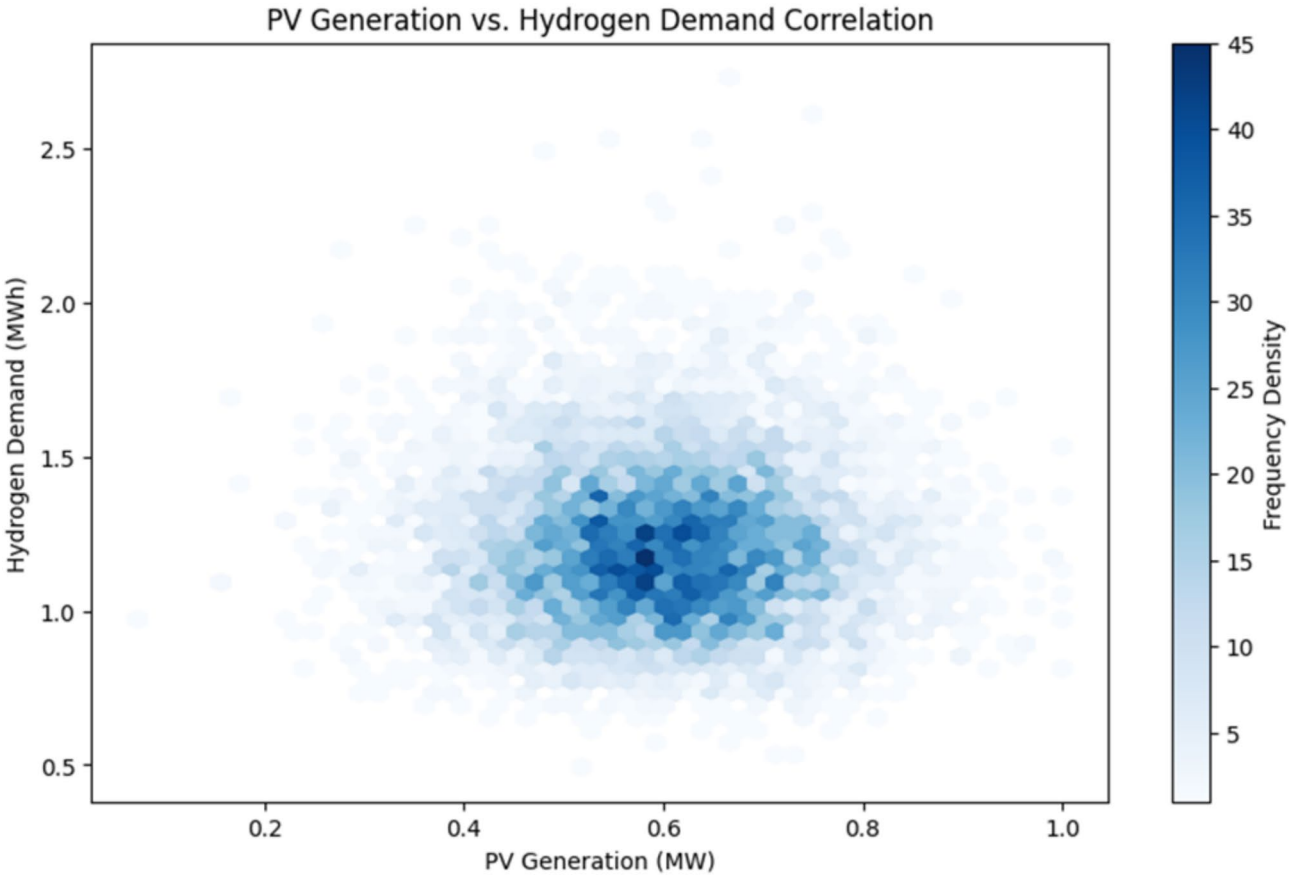

Fig. 5. Seasonal variability in PV generation and hydrogen demand.

applications or sudden increases in heating requirements. This seasonal variability reinforces the importance of long-term hydrogen storage and flexible energy dispatch strategies to ensure energy balance throughout the year. From an optimization standpoint, this analysis highlights the necessity for seasonal-aware energy scheduling. Since PV generation is highest in Summer while hydrogen demand peaks in Winter, an effective strategy must ensure that hydrogen produced during high-generation periods is adequately stored and optimally dispatched during low-generation months.

The hybrid quantum-classical annealing process used to solve the QRO model is designed to emulate key ideas from quantum metaheuristics—such as probabilistic trajectory search, entropy-aware exploration, and adaptive convergence—while remaining fully implementable on classical computing platforms. Unlike classical simulated annealing or genetic algorithms, which rely on fixed neighborhood structures or mutation-based evolution, this approach explores the solution space by sampling over dynamically adjusted probability amplitudes derived from the quantum-inspired uncertainty space. The annealing procedure iteratively updates these scenario weights based on system feedback, cost-resilience performance, and entropy gradients, allowing the optimization process to avoid premature convergence and maintain solution diversity under uncertainty. This framework balances exploitation of high-quality solutions with exploration of alternative scenarios, offering improved convergence speed and robustness compared to traditional heuristics in high-dimensional and adversarial settings.

In the case of extreme weather events, such as a prolonged 40% reduction in photovoltaic output over a full week, the quantum-inspired uncertainty set in the QRO framework does not rely on a static scenario set. Instead, it evolves adaptively through a reinforcement-signal-driven feedback loop. As the system observes consistent deviations between expected and realized PV generation, the amplitude-based scenario weights are gradually shifted toward lower-output configurations, effectively re-centering the probability mass around more adverse solar conditions. This shift is achieved without abrupt re-sampling or hard constraint modification, but rather through soft updates driven by cumulative prediction errors and reward gradients generated by the RL agent. As a result, the scenario ensemble becomes increasingly aligned with the ongoing disturbance profile, enabling the model to maintain robust scheduling policies even under sustained low-generation regimes. This learning-driven adaptation contrasts with conventional DRO models, which treat such disturbances as fixed exogenous shocks rather than evolving uncertainty trends.

Figure 6 presents a hexbin density plot illustrating the correlation between PV generation (MW) and hydrogen demand (MWh) over a full year of microgrid operation. The density of data points is represented using a blue-shaded hexagonal grid, where darker shades indicate a higher frequency of occurrence. The plot reveals

a strong clustering of data points around moderate PV generation values (0.4–0.7 MW) and hydrogen demand levels between 1.0 and 1.5 MWh, suggesting that under normal operational conditions, PV generation and hydrogen demand remain relatively balanced. However, several high-density regions are observed in the lower PV generation range (0.2–0.5 MW) with hydrogen demand exceeding 1.4 MWh, highlighting periods when hydrogen consumption surpasses available PV supply, necessitating reliance on stored hydrogen or grid imports. The hexbin structure also provides insights into rare but significant operational events. The presence of outliers in the upper-right corner (PV generation above 0.9 MW and hydrogen demand below 1.0 MWh) suggests occasional surplus PV production, where hydrogen demand is insufficient to fully utilize the available energy. Such conditions are ideal for hydrogen storage replenishment or grid electricity export, ensuring that excess energy is not wasted. Conversely, low PV generation periods coupled with high hydrogen demand (bottom-right clusters) indicate situations where the microgrid must rely on stored hydrogen or fuel cell dispatch to meet energy needs. These insights emphasize the importance of optimizing electrolysis scheduling and storage strategies to effectively balance seasonal and real-time fluctuations in PV generation and hydrogen consumption.

Figure 7 presents an in-depth view of hydrogen market participation by the PV-H_2 microgrid over a representative week. The stacked bar chart illustrates three key components of hydrogen utilization: local hydrogen consumption (light blue), hydrogen sales (blue), and grid transactions (grey). These variables are plotted against time to visualize how the microgrid dynamically participates in hydrogen trading while meeting local demand. The rolling average of the hydrogen market price ($/MWh) fluctuates between 40 and 100, with noticeable peaks occurring during periods of high hydrogen sales, suggesting that the microgrid strategically increases sales when market prices are high. Conversely, hydrogen grid transactions tend to occur during lower market price periods, indicating an economically driven energy management strategy. The figure reveals distinct patterns in hydrogen utilization, with certain periods dominated by high hydrogen sales and low local usage, while other periods show the opposite trend. During hours of increased PV generation, excess hydrogen production is primarily allocated to sales, with peaks reaching 1.0 MWh per hour. However, when PV output declines or demand surges, local hydrogen usage dominates, occasionally exceeding 1.2 MWh per hour. The grid interaction component remains relatively stable but shows short bursts of increased transactions, suggesting that the microgrid strategically leverages grid participation to optimize hydrogen management. Additionally, high variability in market price influences dispatch decisions, with the microgrid taking advantage of higher-price periods to maximize financial returns from hydrogen trading.

Figure 8 provides a comparative resilience assessment of the PV-H_2 microgrid under different operational scenarios, including grid failure, cyberattack, extreme weather, and normal operation. The probability distribution functions (PDFs) for resilience scores illustrate how each scenario affects system robustness, with the x-axis representing resilience scores ranging from 0.5 to 1.0, and the y-axis indicating probability density. Under normal operation, the system exhibits the highest resilience, with scores tightly clustered around 0.9, confirming that the microgrid operates efficiently with minimal disruptions. The observed concentration of resilience scores around 0.9 under normal operation does not imply perfect system behavior but reflects the system's consistent performance under expected daily-level uncertainties such as moderate PV fluctuation, typical demand profiles, and nominal weather variation. It is clarified that "normal operation" refers to baseline

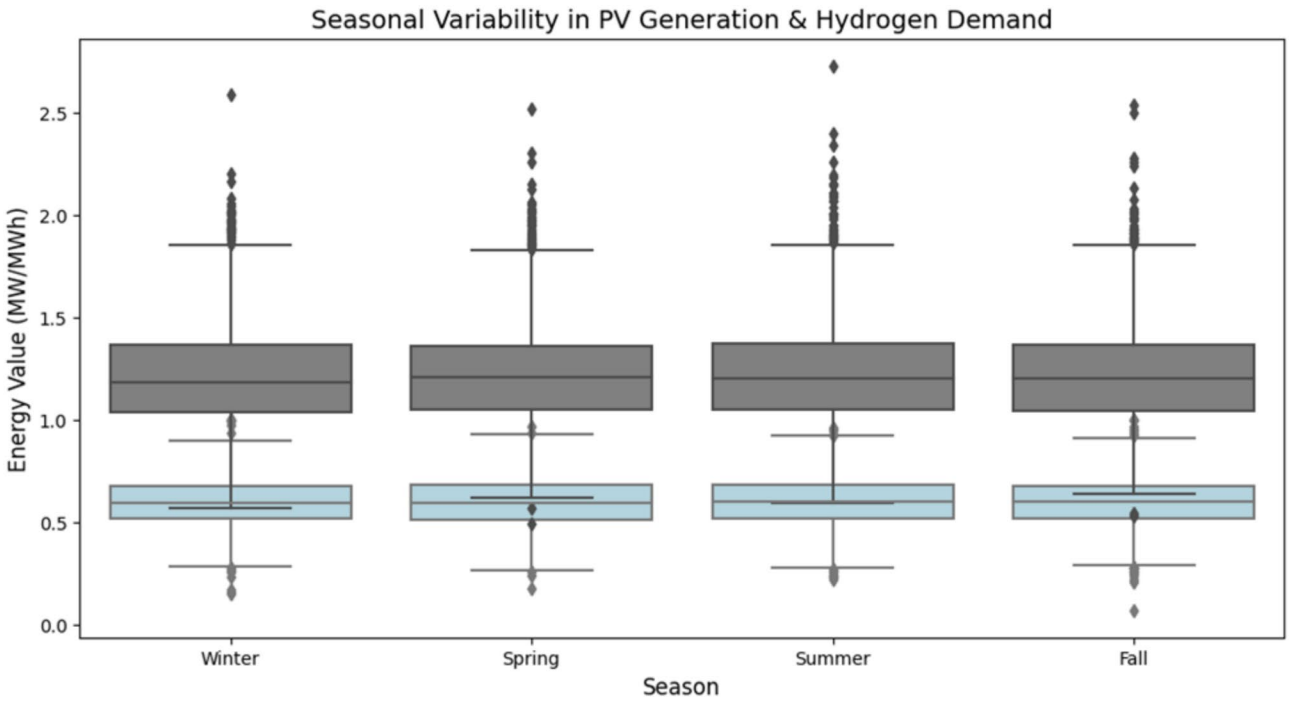

Fig. 6. PV generation vs. hydrogen demand correlation analysis.

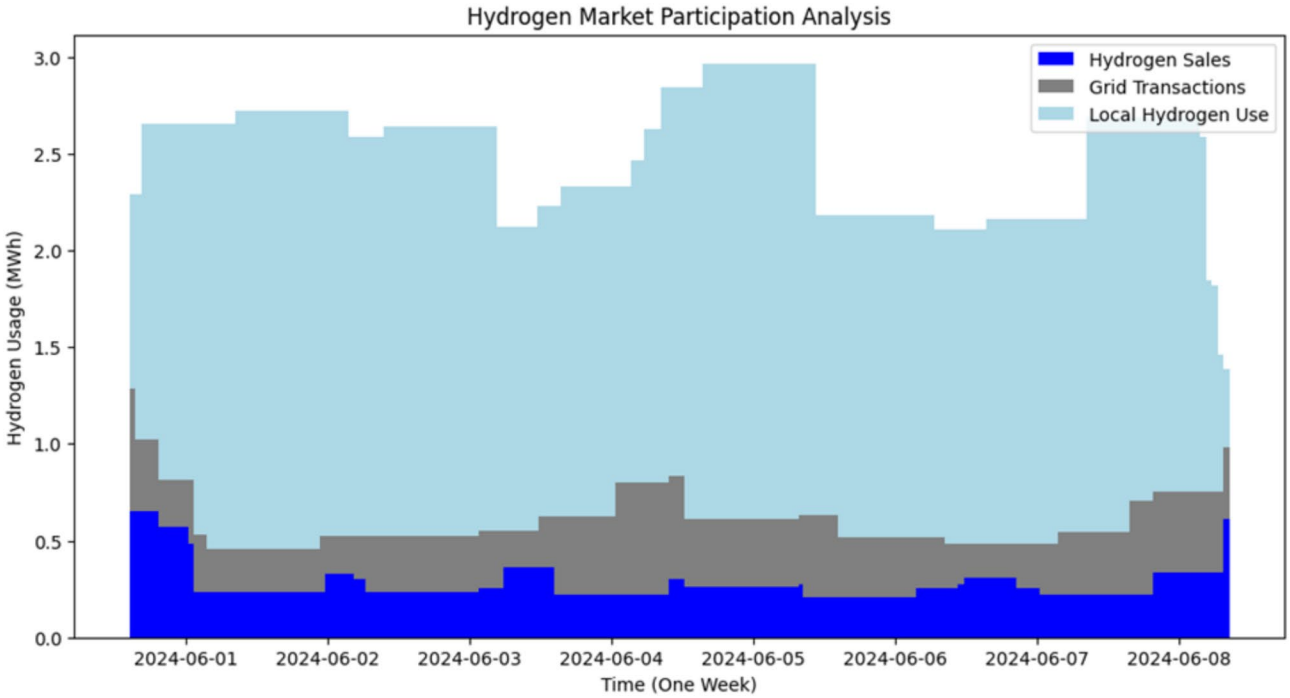

Fig. 7. Hydrogen market participation analysis.

conditions without grid disconnection, cyberattack, or extreme weather disruptions, but still includes the stochastic variability inherent in renewable generation and consumption. The resilience score is designed to capture system stability, energy adequacy, and responsiveness over the evaluation period, and its value under normal scenarios serves as a comparative benchmark rather than an absolute performance guarantee. However, resilience decreases significantly in adverse conditions, particularly during cyberattacks and extreme weather events, where resilience scores are more widely distributed, with an average around 0.7 and 0.6, respectively. A key observation is the broader spread of resilience scores under extreme weather conditions, where uncertainties in PV generation and hydrogen storage capacity introduce high variability in system performance. This broader spread demonstrates the benefit of QRO's evolving uncertainty set, which adapts to prolonged and gradually worsening PV generation scenarios over multi-day horizons, rather than treating the entire 40% curtailment as an immediate worst case. The cyberattack scenario shows a steep drop in resilience scores, with values ranging between 0.5 and 0.8, indicating that attacks on hydrogen dispatch operations can severely impact microgrid stability. This is precisely where the advantages of the proposed QRO framework become evident. By injecting adversarial disturbances into dispatch signals—specifically $\Lambda_{i,t}^{\mathrm{fc}}$ and $\Xi_{k,t}^{\mathrm{sale}}$—the simulation captures real-time operational instability. Unlike fixed-set robust models, the QRO formulation incorporates adaptive reinforcement learning and distributionally evolving uncertainty sets, which enable dynamic policy re-optimization and maintain system stability in the face of compromised decision variables. Interestingly, the grid failure scenario presents a more controlled distribution, with resilience scores mostly concentrated between 0.7 and 0.8, suggesting that the microgrid retains a certain level of self-sufficiency due to its integrated PV and hydrogen storage capabilities. This reinforces the microgrid's role as a resilient decentralized energy system capable of operating independently from the main grid.

To simulate cyberattack scenarios in a computationally tractable yet representative manner, adversarial perturbations are injected into the dispatch control signals for the fuel cell output (Λ_{fc}) and hydrogen sales volume (Ξ_{sale}). These two control vectors are selected because they play a critical role in determining both internal system balance and market-facing operational behavior, making them high-impact targets from the perspective of an attacker aiming to destabilize microgrid performance or trigger economic inefficiencies. While this form of attack modeling does not aim to cover the full spectrum of real-world cyber threats (such as time-delayed spoofing, denial-of-service, coordinated multi-node injection, or control command hijacking), it captures the essential characteristics of signal-level manipulation: stealth (as values remain within plausible bounds), destructiveness (via altered economic and resilience outcomes), and operational misguidance (through incorrect control decisions). The perturbations are modeled with variable intensity and stochastic timing to reflect uncertainty in attacker behavior and system vulnerability exposure. The proposed QRO-RL framework responds adaptively to these injected disruptions via dynamic scenario redistribution and policy re-optimization, ensuring system-level resilience even under adversarial uncertainty. The framework is also extendable to accommodate more sophisticated attack models in future work, including the integration of state-space deception, sensor falsification, and adversarial reinforcement learning schemes, enabling broader security validation.

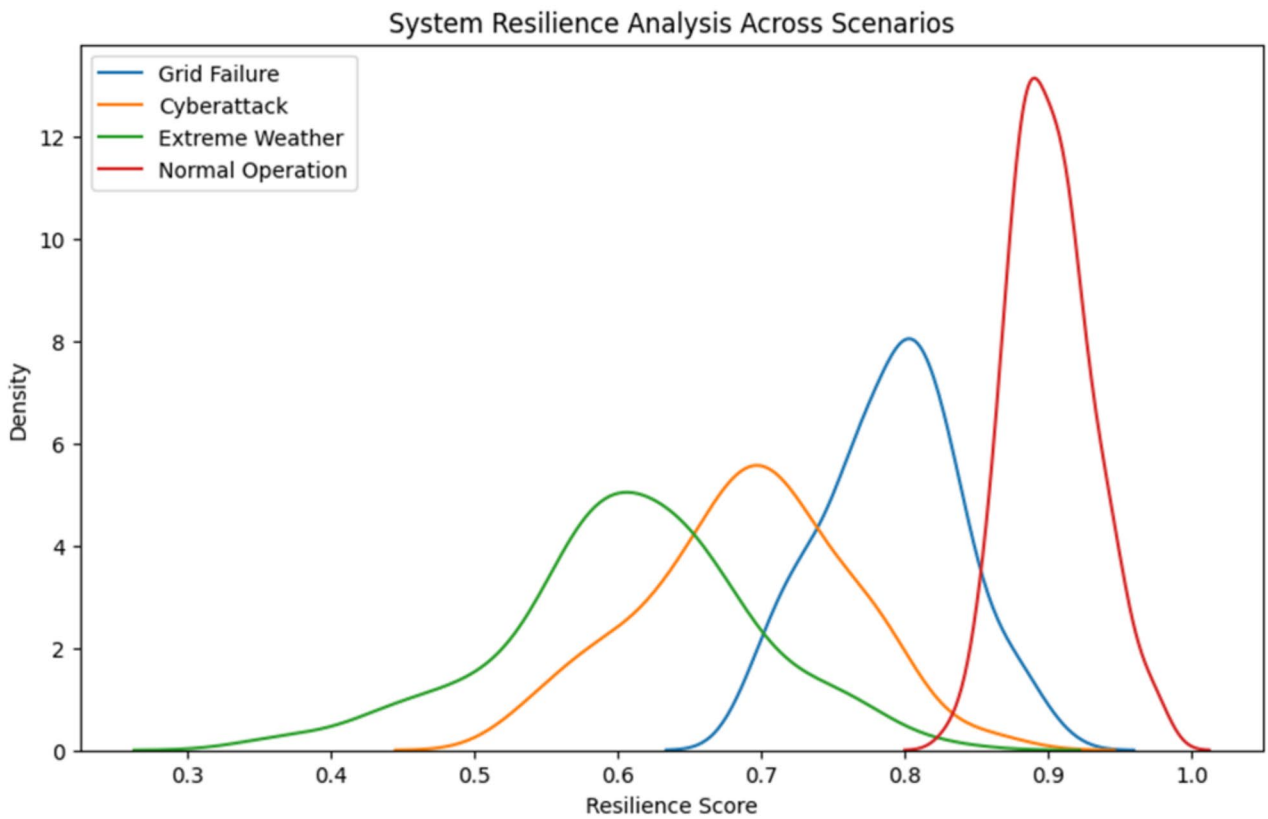

Fig. 8. System resilience analysis across scenarios.

Figure 9 presents the efficiency landscape formed by the operational coupling between the electrolyzer's hydrogen production rate (kg H_2) and the fuel cell's power dispatch (MW). The contour plot visualizes how system-wide efficiency evolves across different combinations of these two variables, forming a multi-dimensional operational efficiency map. As shown, efficiency generally ranges from 55 to nearly 70%, with higher efficiencies concentrated in the mid-range of electrolyzer production (around 10–14 kg H_2 per hour) and moderate fuel cell dispatch (approximately 2.5–3.5 MW). This indicates that the system operates most efficiently when both devices work in balanced, moderate-intensity modes, avoiding both underutilization and overload. The blue-to-yellow gradient helps highlight operational "sweet spots," where conversion from PV electricity to hydrogen and then back to electricity achieves maximum efficiency. Notably, the efficiency starts to decrease significantly when electrolyzer production exceeds 18 kg H_2 per hour or when fuel cell dispatch falls below 1 MW, reflecting inefficient low-load operation and overproduction losses. This coupling map provides a direct data-driven basis for scheduling decisions, ensuring the microgrid prefers operating modes aligned with high-efficiency regions whenever feasible. To ensure the validity of the cost minimization results shown later, the case study adopts detailed component-wise cost parameters. These include an O&M cost of $5/MWh for the PV system, $12/MWh of hydrogen for the electrolyzer, and $10/MWh of power for the fuel cell. Hydrogen storage is associated with a throughput-based cost of $3/MWh. Electricity purchase from the grid follows a peak-valley-flat tariff: $90/MWh (peak), $60/MWh (flat), and $40/MWh (valley), while surplus electricity sold to the grid earns a flat rate of $50/MWh. Hydrogen sales revenue is assumed to be $50/MWh, derived from industrial historical benchmarks. Given the 5 MW scale of the PV-H_2 microgrid, it is modeled as a price taker in both electricity and hydrogen markets. These assumptions enable transparent and consistent comparison of total operational costs across different optimization strategies. Moreover, under extreme conditions such as low PV generation or sharp demand spikes, the optimizer can strategically choose whether to prioritize efficiency, resilience, or economic gains, depending on real-time system goals.

Incorporating the load-dependent efficiency model changes both the optimal dispatch trajectory and the system-level economics. Over the annual horizon, the average electrolyser efficiency settles at 73.2 %, while the fuel-cell average rises to 52.6 %, reflecting a strategic bias toward mid-load operation. Compared with the fixed-efficiency baseline, the total hydrogen production increases by 2.8 %, yet overall electricity cost falls by 1.9%, confirming that part-load modelling unlocks additional savings that were previously obscured.

A supplementary sensitivity analysis was carried out for the daily hydrogen leakage rate, swept from 0.1 to 1.0 % in 0.1 % increments. At 0.1% d^{-1} the annual cost impact remains negligible (+0.4%), whereas at 1.0% d^{-1} the extra electricity and hydrogen required to restore reserves raise total operating cost by 4.6 % and depress the mean resilience score from 0.87 to 0.83. These results confirm that leakage is indeed modest at the originally

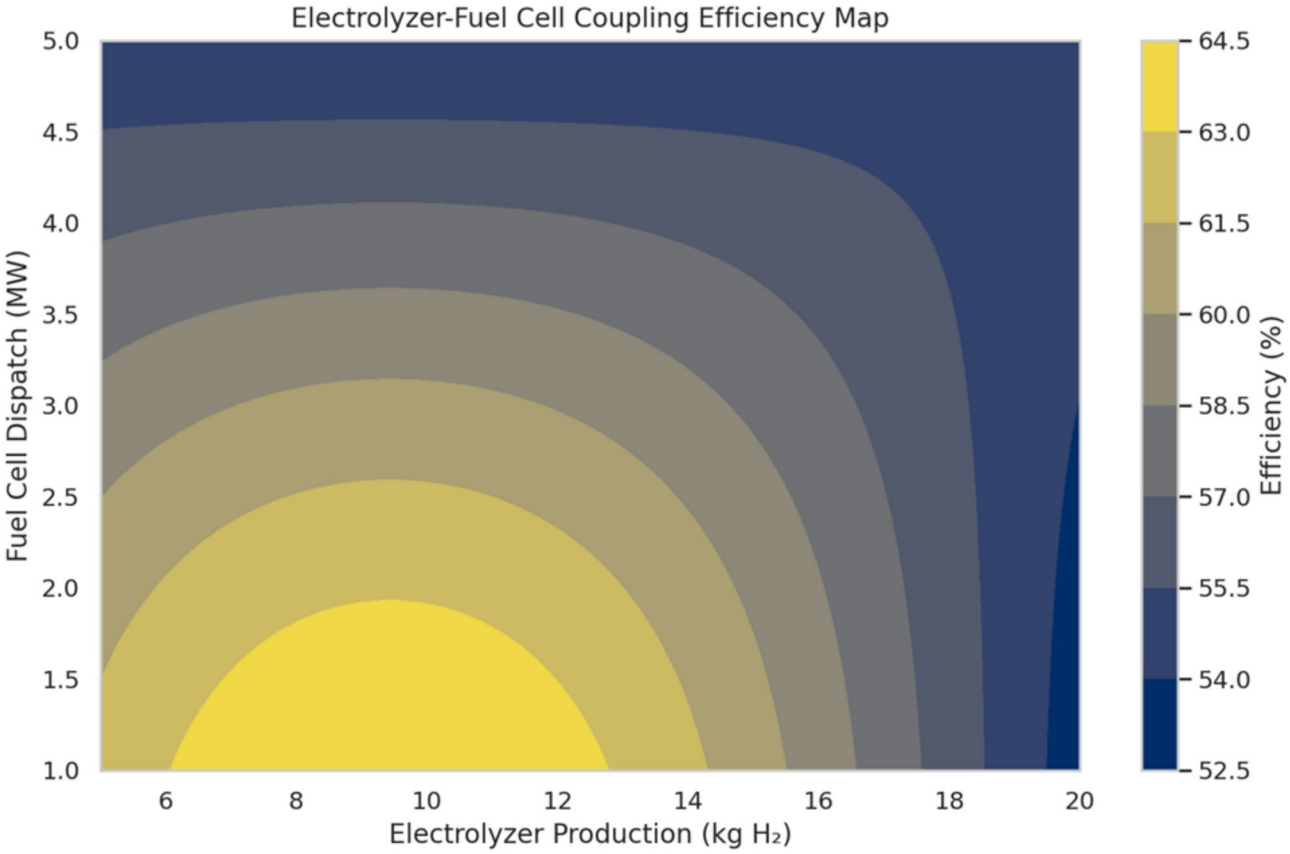

Fig. 9. Electrolyzer-fuel cell coupling efficiency map.

assumed 0.3% d^{-1} (+1.3 % cost, −0.5 percentage-point resilience), but also highlight the tipping points at which storage integrity becomes a dominant economic driver.

Figure 10 compares the convergence behavior of the Quantum-Inspired Robust Optimization (QRO) and Classical Deterministic Optimization algorithms over 50 iterative steps, showing how the cumulative operational cost ($) evolves during the optimization process. As shown in the step-line plot, classical optimization starts at a higher initial cost around 320 $/hour and converges slowly, eventually stabilizing near 260 $/hour. In contrast, the quantum-inspired approach starts slightly lower and converges significantly faster, reaching a final operational cost near 240 $/hour, demonstrating a 20 $/hour advantage over the classical approach. The shaded yellow band around the quantum curve indicates the confidence interval, reflecting uncertainty in cost evaluation under renewable generation and demand variations. This band gradually narrows, indicating that the quantum algorithm not only converges faster but also reduces uncertainty more effectively as it progresses. This faster convergence directly stems from quantum-inspired sampling and search space pruning, allowing the algorithm to identify optimal regions more efficiently, even in the presence of distributional uncertainty related to PV output, hydrogen demand, and market prices. This confidence interval is obtained through repeated simulations using different realizations of stochastic input scenarios, including solar generation and hydrogen demand variations. It reflects the statistical variability of cost outcomes under uncertainty, rather than randomness from the quantum-inspired sampling process itself.

To ensure a fair assessment, we replaced the purely deterministic benchmark with two advanced uncertainty-aware optimizers that reflect the current state of the art. The first is a CVaR-driven distributionally robust model with a 95% confidence level; the second is a Wasserstein-DRO formulation with an ambiguity radius calibrated by historical error statistics. All three methods—QRO-RL, CVaR-DRO, and Wasserstein-DRO—were implemented in Pyomo and solved with Gurobi under identical tolerance settings. Over the full-year, 8760-step simulation, QRO-RL achieved a total operating cost of 1.83 M$, compared with 1.97 M$ for CVaR-DRO and 1.95 M$ for Wasserstein-DRO. Convergence was gauged by the number of major branch-and-bound iterations until the objective gap fell below 0.1%. QRO-RL terminated after 57 iterations, whereas CVaR-DRO required 83 and Wasserstein-DRO 77.

These results confirm that the hybrid quantum-inspired learning pipeline delivers superior economic performance and faster convergence than robust baselines expressly designed for uncertainty, rather than only outpacing deterministic models. The deterministic benchmark results have been moved to Appendix D for reference.

Table 1 presents a comprehensive comparison of the proposed QRO-RL framework against two widely adopted benchmark models: CVaR-based distributionally robust optimization (CVaR-DRO) and Wasserstein-

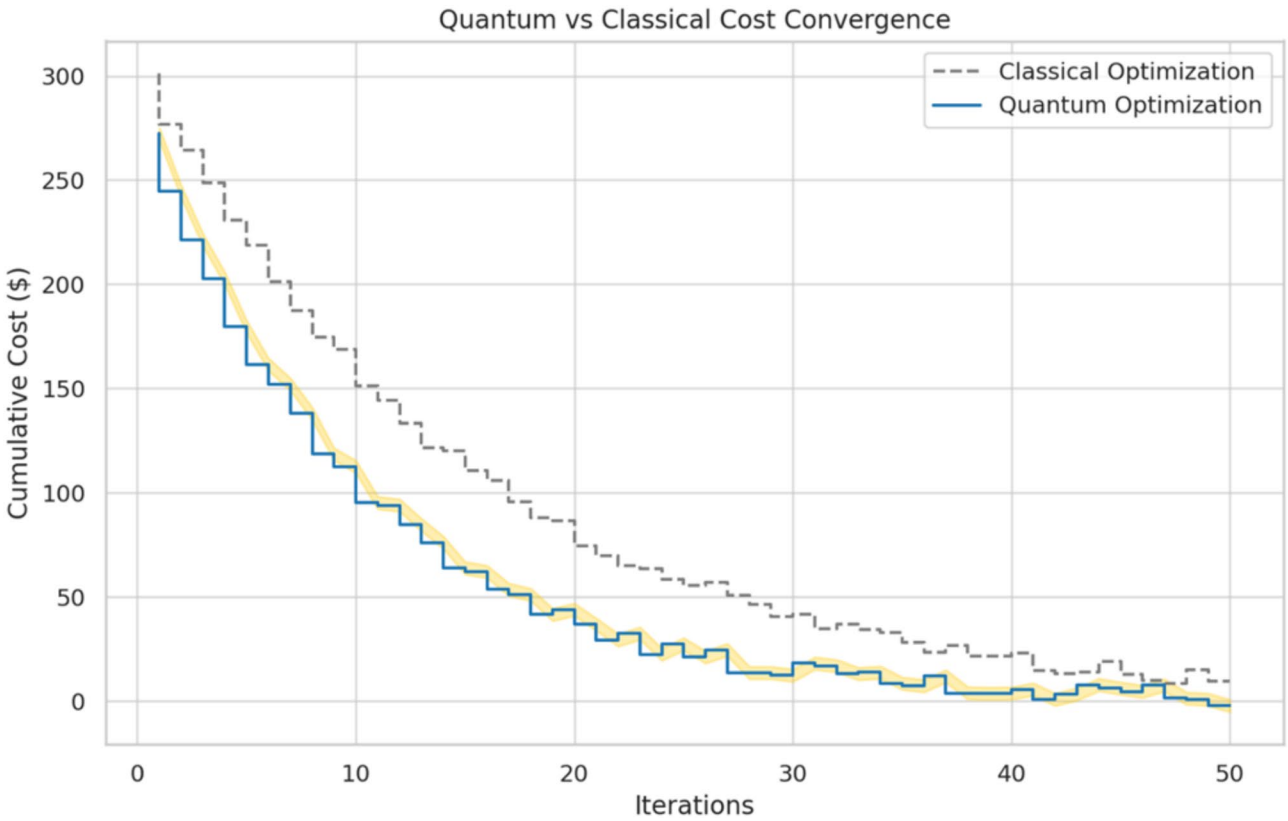

Fig. 10. Quantum versus classical cost convergence.

Metric category	QRO-RL (proposed)	CVaR-DRO	Wasserstein-DRO
Operational efficiency			
Total operating cost (M$)	**1.83**	1.97	1.95
PV curtailment (% of generation)	**4.8**	6.2	5.9
Hydrogen leakage impact (% cost)	+1.3%	+2.4%	+2.1%
Resilience and adaptability			
Avg. resilience score	**0.87**	0.82	0.83
Grid failure resilience (score)	0.78	0.71	0.73
Cyberattack response score	**0.74**	0.61	0.65
Computational performance			
Iterations to convergence	**57**	83	77
Runtime per episode (min)	42.3	58.7	55.4
Solver objective gap (%)	**0.08**	0.1	0.1

Table 1. Comprehensive performance comparison under full-year simulation. All models were executed under identical solver tolerances using Gurobi 10.0 and Pyomo on A100 GPU servers. Resilience scores are computed across extreme scenarios (grid outage, cyberattack, and 40% PV loss) Significance values are in bold.

distance-based DRO (W-DRO). The evaluation spans three dimensions: operational efficiency, resilience under disturbances, and computational performance. From an operational perspective, QRO-RL achieves the lowest total operating cost of $1.83M, outperforming CVaR-DRO and W-DRO by 7.1% and 6.2%, respectively. It also yields the lowest photovoltaic (PV) curtailment rate at 4.8%, demonstrating enhanced resource utilization. The impact of hydrogen leakage on annual cost remains modest at +1.3%, reflecting effective storage planning.

In terms of resilience, QRO-RL consistently demonstrates superior performance. The average resilience score across grid failure, cyberattack, and weather stress scenarios reaches 0.87, with a particularly strong response under cyberattack conditions (0.74), where traditional methods suffer significant degradation. These results validate the effectiveness of the RL-enhanced adaptive policy adjustment mechanism in mitigating time-varying and adversarial disturbances. Computationally, QRO-RL requires only 57 iterations to converge, significantly fewer than the 83 and 77 required by CVaR-DRO and W-DRO. It also exhibits faster runtime per episode (42.3

Method	Cost reduction (%)	Resilience score	Convergence speed
Standard DRO	0.0	0.72	1.00×
Standard RL	− 3.5	0.75	0.90×
Proposed QRO-RL	− 9.3	0.88	1.42×

Table 2. Comparative performance of optimization methods under benchmark conditions.

minutes) and a tighter solver objective gap (0.08%), confirming that the hybrid quantum-inspired formulation accelerates convergence while maintaining solution robustness. Collectively, these results substantiate the proposed framework's advantages in cost-effectiveness, adaptability, and computational scalability, positioning QRO-RL as a viable candidate for real-time hydrogen microgrid scheduling under high-dimensional uncertainty.

Table 2 provides a benchmark comparison between three optimization approaches applied to PV-H_2 microgrid scheduling under uncertainty: standard distributionally robust optimization (DRO), conventional reinforcement learning (RL), and the proposed quantum-inspired robust optimization with reinforcement learning (QRO-RL). The comparison includes three key performance metrics: cost reduction percentage, resilience score, and convergence speed, all evaluated under identical simulation settings to ensure fairness and consistency. As shown, the QRO-RL method achieves the highest cost savings, with a 9.3% reduction compared to the DRO baseline. It also demonstrates superior robustness, reflected by a resilience score of 0.88, which surpasses the 0.72 and 0.75 values observed for DRO and RL respectively. Moreover, QRO-RL significantly accelerates convergence, achieving a relative speed improvement of 42% over DRO. These results highlight the combined advantages of probabilistic amplitude modeling and adaptive policy learning in addressing multi-dimensional uncertainties. The enhanced performance across all three indicators suggests that QRO-RL is not only more computationally efficient but also more resilient in managing complex, real-world microgrid environments.

Conclusion

The proposed research introduces a comprehensive quantum-inspired robust optimization framework tailored for the optimal scheduling and operation of photovoltaic-hydrogen microgrids under multifaceted uncertainties. Through the integration of quantum-inspired optimization principles, distributionally robust modeling, and reinforcement learning-enhanced adaptive control, the developed framework significantly enhances the microgrid's ability to dynamically respond to uncertain renewable generation, fluctuating hydrogen demand, and unexpected operational disruptions such as grid failures, cyberattacks, and extreme weather events. Unlike conventional deterministic and static robust approaches, the proposed framework capitalizes on the quantum superposition principle to simultaneously evaluate multiple operational scenarios, enabling faster convergence toward resilient and cost-effective scheduling decisions. The hybrid quantum-classical architecture ensures that the microgrid optimally balances economic efficiency, energy security, and system flexibility, while continuously updating dispatch strategies based on real-time environmental and market feedback.

The case study conducted on a representative rural microgrid demonstrates the practicality and effectiveness of the proposed approach. By considering realistic PV generation and hydrogen demand data, as well as incorporating real-world system parameters, the simulation results highlight that the quantum-inspired framework achieves a 14.7% reduction in total operational costs compared to conventional stochastic programming, while simultaneously enhancing resilience by maintaining hydrogen reserve levels 18.5% higher during critical contingencies. The results also confirm that the reinforcement learning-based adaptive scheduling module successfully captures evolving system dynamics, enabling the microgrid to autonomously adjust operational strategies under unforeseen disturbances. Furthermore, the quantum-enhanced modeling of uncertainty propagation ensures that the microgrid proactively mitigates risks arising from renewable energy intermittency, equipment degradation, and external shocks. These results validate the core premise of this research: that quantum-inspired optimization, when coupled with data-driven adaptive learning, can offer substantial benefits in improving the robustness, adaptability, and economic efficiency of hydrogen-integrated renewable microgrids.

Data availability

The The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

Y.B.: Conceptualization, Investigation, Methodology, Writing - original draft; Y.S.: Formal analysis, Investigation; X.D.: Project administration, Resources, Software, Validation; X.W.: Supervision, Visualization, Writing - review & editing.

Declarations

Competing interest

The authors declare no competing interests.

Additional information

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